

Term Project

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1 Introduction

In this project, we will explore two different datasets using multiple linear regression, ridge regression, lasso regression, transformed regression (specifically Sqrt, Log1p, and Box-Cox), second-order polynomial regression including interaction terms, and neural networks (two, three, and four layers). The two datasets we will be using are:

- **Insurance Charges:** This dataset contains information about different primary beneficiaries of health insurance. The goal is to predict the primary beneficiaries insurance charges given their other characteristics. The dataset can be found at <https://www.kaggle.com/datasets/mirichoi0218/insurance>.
- **Insurance Annual Premium:** This dataset contains information about different primary beneficiaries of health insurance. The goal is to predict the primary beneficiaries annual premium given their other characteristics. The dataset can be found at <https://www.kaggle.com/datasets/mohankrishnathalla/medical-insurance-cost-prediction>.

2 Hyperparameter Tuning

To optimize the regularization hyperparameter for Ridge and Lasso regression, we employed a three-stage grid search, using 5-fold cross-validation to maximize the mean R^2 score.

- Stage 1: A broad logarithmic search to find the general neighborhood of the best hyperparameter.
- Stage 2: An intermediate refinement around the Stage 1 winner.
- Stage 3: A high-resolution linear search to finalize the value based on the Stage 2 results.

For the second-order polynomial regression model, we used ridge regression to prevent overfitting. To optimize the power transformation lambda for Box-Cox, we employed a grid search of standard power transformations to determine which lambda yields the highest mean R^2 via 5-fold cross-validation.

We optimized the neural network's activation functions, hidden layer sizes, and learning rate using a sequential grid search, maximizing the mean R^2 via 5-fold cross-validation. First, activation functions were tuned layer-by-layer, from the initial hidden layer through to the output. We then determined the hidden layer sizes; for two-layer architectures, we constrained the second layer to be less than or equal to the first. Finally, the learning rate was tuned.

3 Insurance Charges

3.1 Exploratory Data Analysis

The Insurance Charges dataset contains 1338 observations each having 7 variables. These variables are:

- age: The age of the primary beneficiary.
- sex: The insurance contractor sex: female or male.
- bmi: The body mass index of the primary beneficiary.
- children: The number of children/dependents covered by health insurance
- smoker: If the primary beneficiary smokes.

- region: The beneficiary's residential area in the US: northeast, southeast, southwest, or northwest.
- charges: (The target variable) Individual medical costs billed by health insurance.

The Insurance Charges dataset is fully populated, i.e. there are no missing values, so we do not exclude any from the analysis. For modeling purposes, we appended an intercept column of ones and converted the categorical sex, smoker, and region variables into dummy variables (sex_male, smoker_yes, region_northwest, region_southeast, region_southwest), omitting sex_female, smoker_no, and region_northeast to prevent multi-collinearity.

Univariate distributions are visualized in Figure 1 (histograms/count-plots) and Figure 2 (box plots for continuous variables). The visualizations reveal that sex and region are approximately uniformly distributed. While age also follows a generally uniform pattern, there is a notable spike in frequency for individuals aged 18 and 19. Additionally, bmi closely follows a normal distribution, whereas children and charges exhibit distributions characteristic of a geometric decay. Notably, both bmi and charges contain several outliers, with the latter displaying a significantly higher concentration. Finally, non-smokers represent the most prevalent group in the dataset.

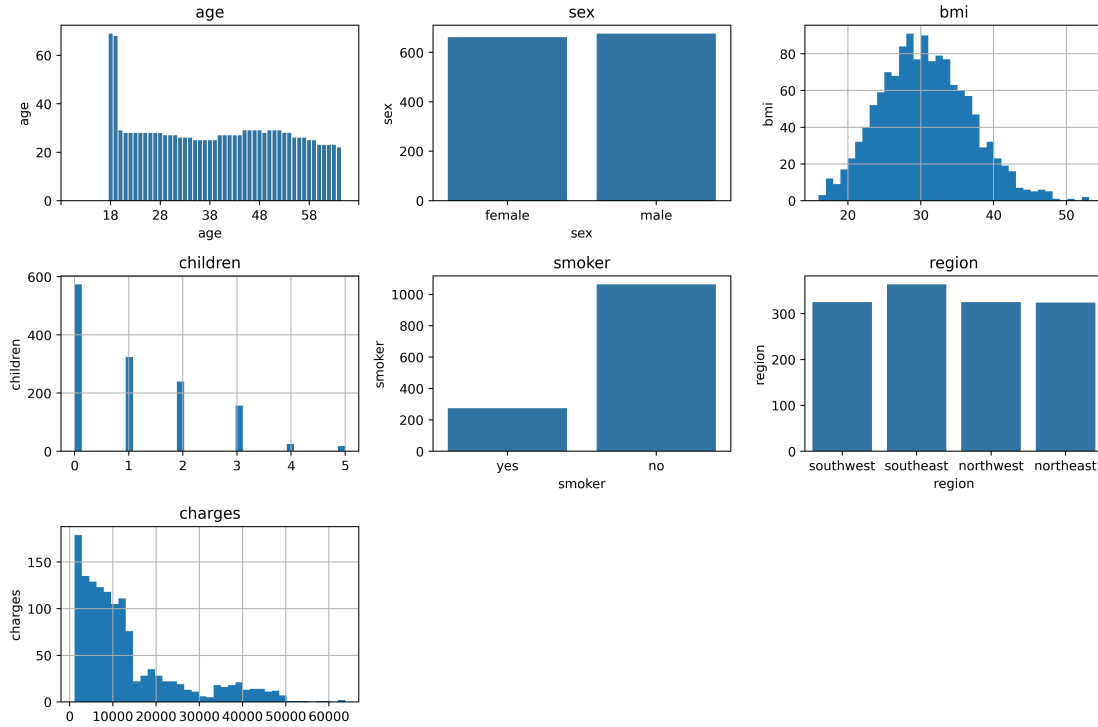


Figure 1: Insurance Charges Histograms and Count-Plots

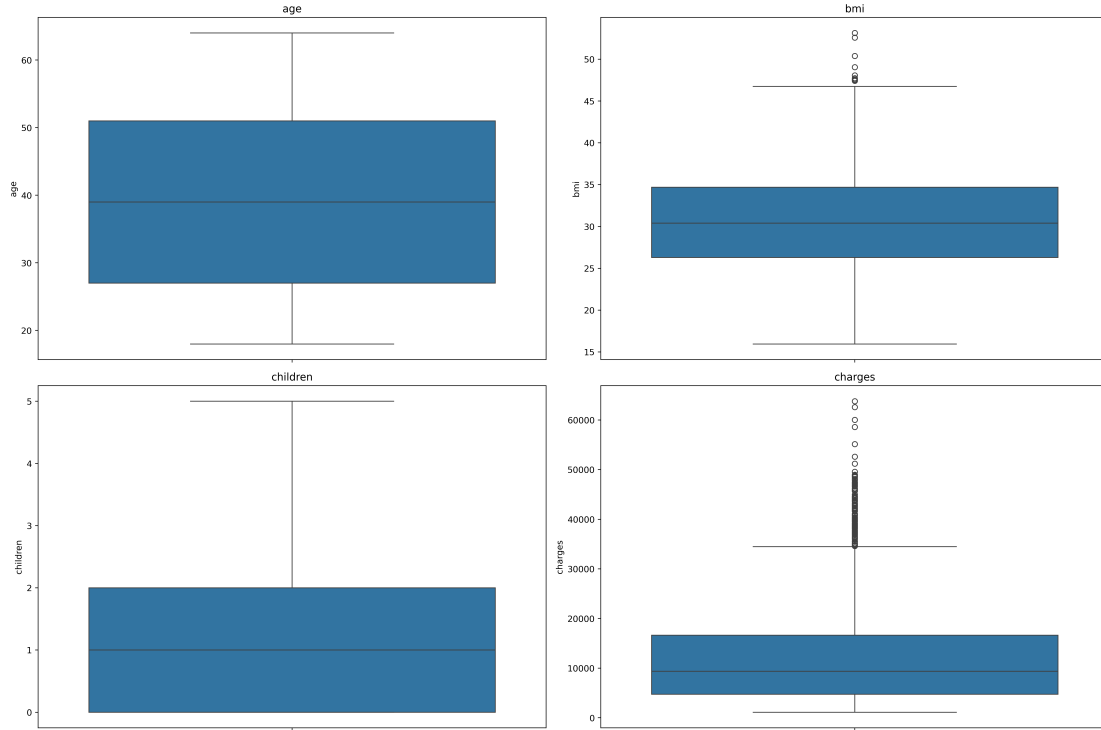


Figure 2: Insurance Charges Box Plots

Figure 3 displays the bivariate relationships between each predictor and the target variable, charges, while the corresponding correlation matrix is provided in Figure 4. Analysis of these figures indicates that the smoker variable has the strongest correlation with charges. In the scatterplot for age, the data points appear to bifurcate into three distinct clusters, each forming its own linear trend. This suggests that latent variables or interaction effects (currently unobserved) might differentiate these groups, as each subgroup shows a strong individual correlation with charges. Furthermore, the figures reveal that the remaining predictor variables exhibit no discernible relationship with charges, appearing as randomly distributed clusters with no clear linear or non-linear patterns. Finally, there is no evidence of significant multi-collinearity or inter-correlation between any of the predictor variables.

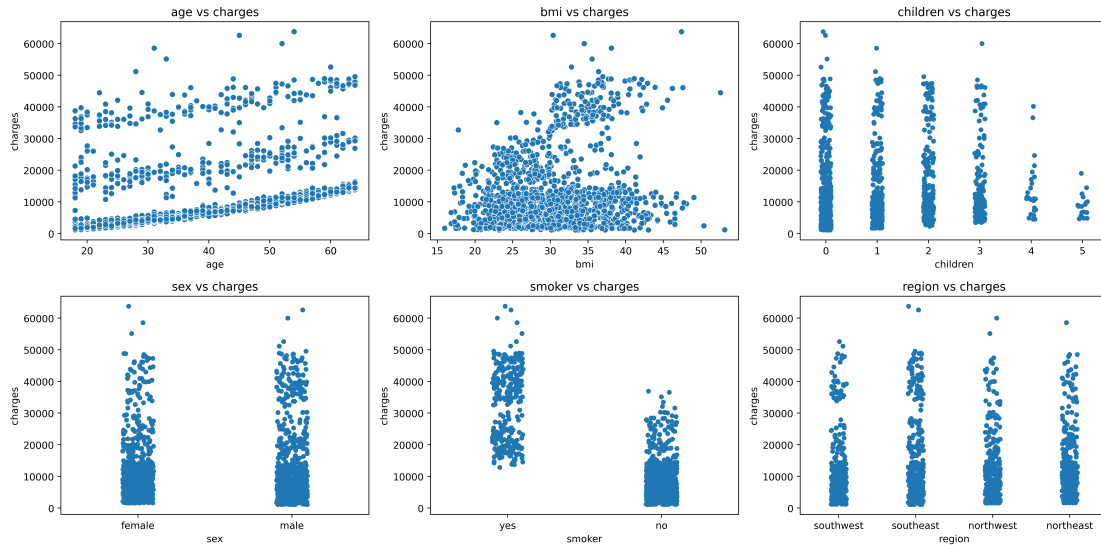


Figure 3: Insurance Charges Scatterplots

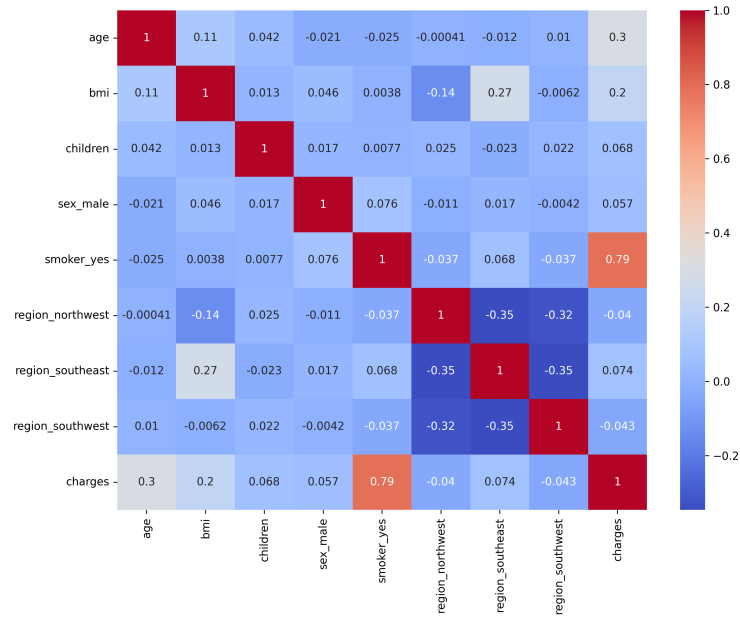


Figure 4: Insurance Charges Correlation Heat Map

Figure 5 illustrates the bivariate relationships among the predictor variables, with data points colored by the target variable, charges. Given the high proportion of categorical predictors, many of these visualizations result in overlapping clusters, making it difficult to definitively determine if most interaction terms would improve model performance. A primary exception is the interaction between bmi and smoker, where it is very clear that the interaction is beneficial. In contrast, for variables such as children and sex, the influence on the target variable remains obscured by the data's density, leaving their potential for interaction-based improvement unclear.

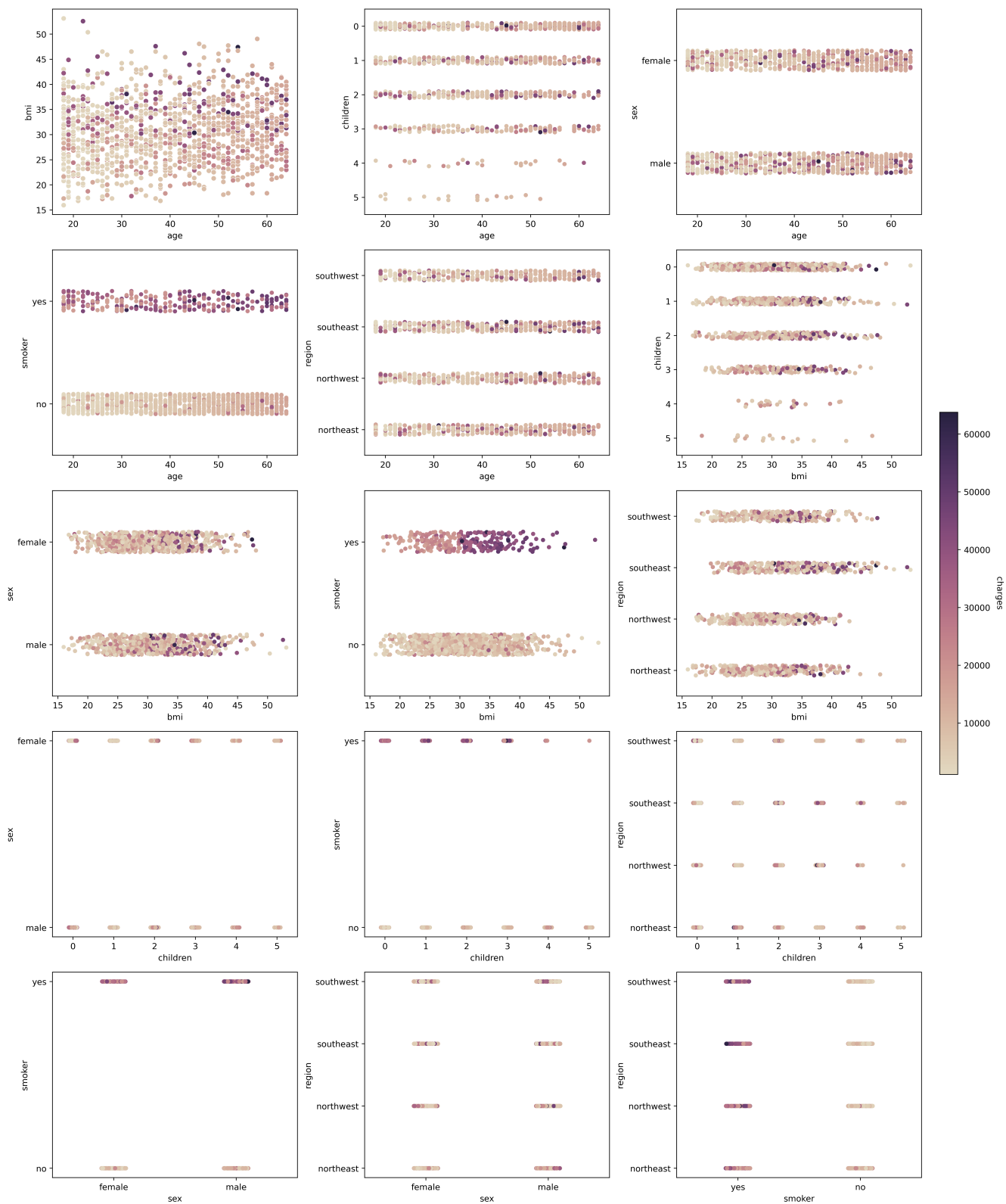


Figure 5: Insurance Charges Hue Plots

3.2 Multiple Linear Regression

3.2.1 Quality of Fit

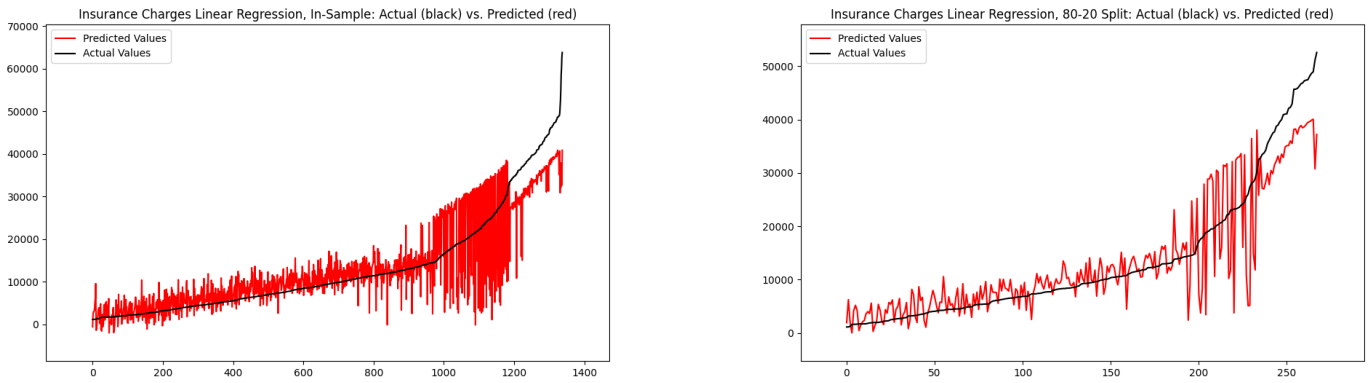


Figure 6: Insurance Charges Linear Regression Plots
Left: In-Sample, Right: Out-of-Sample
black: actual y-values, red: predicted y-values

Table 1: Insurance Charges Linear Regression

Metric	In-Sample	80-20 Split
rSq	0.7509	0.8
rSqBar	0.7494	0.7938
sst	1.961e+11	4.265e+10
sse	4.884e+10	8.53e+09
sde	6062	5739
mse0	3.65e+07	3.183e+07
rmse	6042	5642
mae	4171	3933
smape	37.81	35.02
m	1338	268
dfr	9	9
df	1329	259
fStat	445.2	115.1
aic	2.332e+04	4648
bic	2.336e+04	4680

Table 2: Insurance Charges Linear Regression CV

Metric	num folds	min	max	mean	stdev
rSq	5	0.6525	0.8	0.7401	0.05195
rSqBar	5	0.6418	0.7938	0.7321	0.05356
sst	5	2.986e+10	4.265e+10	3.902e+10	4.668e+09
sse	5	8.53e+09	1.192e+10	9.937e+09	1.194e+09
sde	5	5739	6798	6188	372.4
mse0	5	3.183e+07	4.465e+07	3.714e+07	4.509e+06
rmse	5	5642	6682	6083	365.9
mae	5	3933	4559	4206	212.5
smape	5	35.02	41.32	38.03	2.113
m	5	267	268	267.6	0.4899
dfr	5	9	9	9	0
df	5	258	259	258.6	0.4899
fStat	5	54.05	115.1	86.05	21.39
aic	5	4648	4721	4680	26.77
bic	5	4680	4753	4713	26.76

3.2.2 Feature Selection

Forward Selection: Null, smoker yes, age, bmi, region southwest, region northwest, region southeast, children, sex male

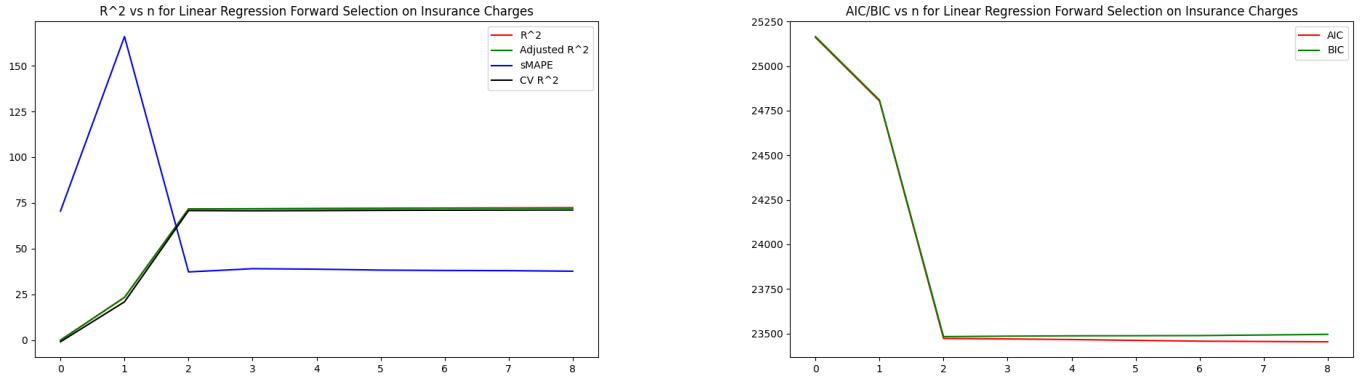


Figure 7: Insurance Charges Linear Regression Forward Selection Quality of Fit Metrics
Left: R^2 , Right: AIC/BIC

Backward Elimination Reversed: Null, smoker yes, age, bmi, region southwest, region northwest, region southeast, children, sex male

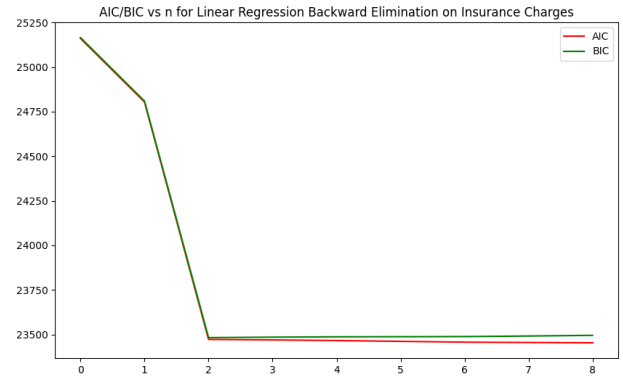
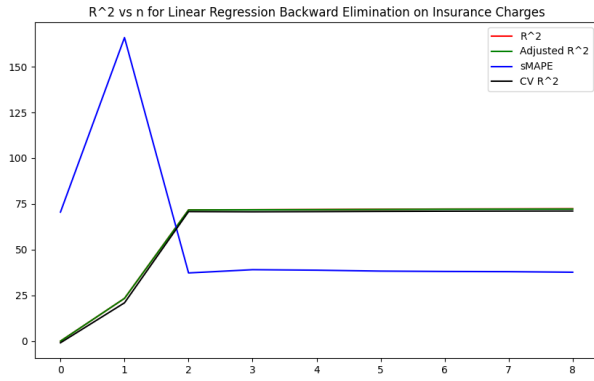


Figure 8: Insurance Charges Linear Regression Backward Elimination Quality of Fit Metrics
Left: R^2 , Right: AIC/BIC

Stepwise Selection: Null, smoker yes, age, bmi, region southwest, region northwest, region southeast, children, sex male

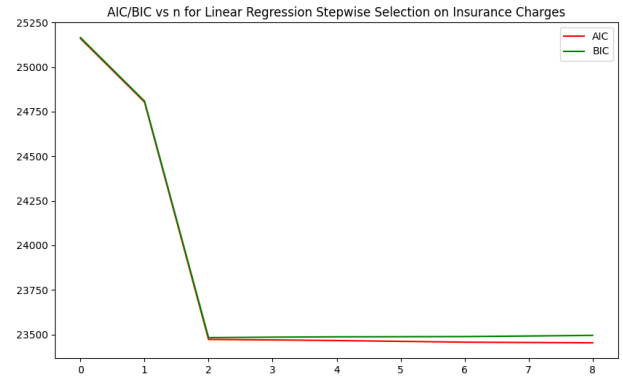
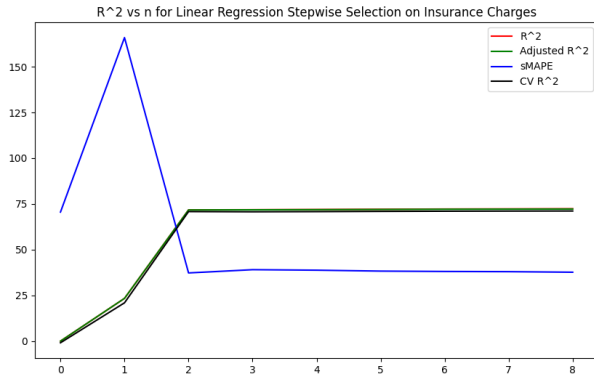


Figure 9: Insurance Charges Linear Regression Stepwise Selection Quality of Fit Metrics
Left: R^2 , Right: AIC/BIC

3.3 Ridge Regression

3.3.1 Quality of Fit

Insurance Charges Ridge alpha Used: 0.009000000000000001

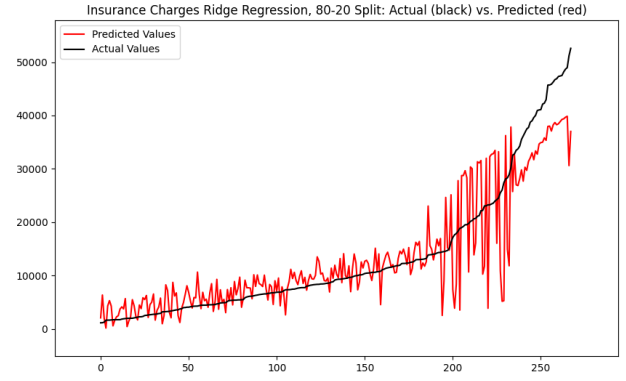
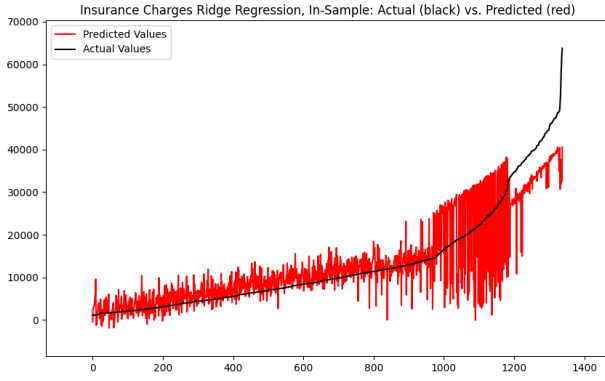


Figure 10: Insurance Charges Ridge Regression Plots
 Left: In-Sample, Right: Out-of-Sample
 black: actual y-values, red: predicted y-values

Table 3: Insurance Charges Ridge Regression

Metric	In-Sample	80-20 Split
rSq	0.7509	0.7993
rSqBar	0.7495	0.7939
sst	1.961e+11	4.265e+10
sse	4.885e+10	8.561e+09
sde	6061	5738
mse0	3.651e+07	3.194e+07
rmse	6042	5652
mae	4182	3951
smape	37.68	34.84
m	1338	268
dfr	8	8
df	1330	260
fStat	501	129.4
aic	2.331e+04	4647
bic	2.336e+04	4676

Table 4: Insurance Charges Ridge Regression CV

Metric	num folds	min	max	mean	stdev
rSq	5	0.6543	0.7993	0.7402	0.05105
rSqBar	5	0.645	0.7939	0.7332	0.05242
sst	5	2.986e+10	4.265e+10	3.902e+10	4.668e+09
sse	5	8.561e+09	1.191e+10	9.938e+09	1.173e+09
sde	5	5738	6780	6177	365
mse0	5	3.194e+07	4.459e+07	3.714e+07	4.432e+06
rmse	5	5652	6678	6084	359.4
mae	5	3951	4562	4217	207.7
smape	5	34.84	41.02	37.88	2.054
m	5	267	268	267.6	0.4899
dfr	5	8	8	8	0
df	5	259	260	259.6	0.4899
fStat	5	61.5	129.4	97.06	23.77
aic	5	4647	4719	4678	26.08
bic	5	4676	4747	4707	26.07

3.3.2 Feature Selection

Forward Selection: Null, smoker yes, age, bmi, children, region southeast, region southwest, region northwest, sex male

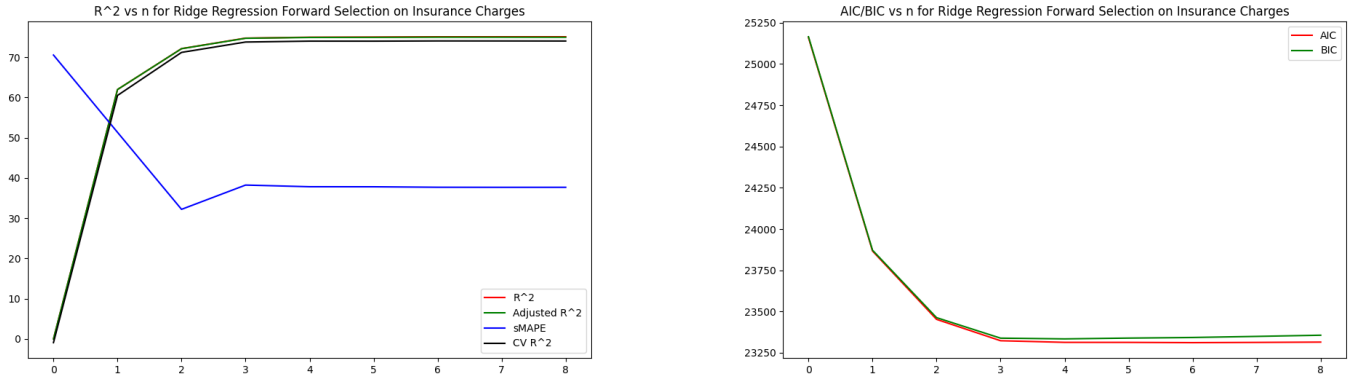


Figure 11: Insurance Charges Ridge Regression Forward Selection Quality of Fit Metrics
Left: R^2 , Right: AIC/BIC

Backward Elimination Reversed: Null, smoker yes, age, bmi, children, region southeast, region southwest, region northwest, sex male

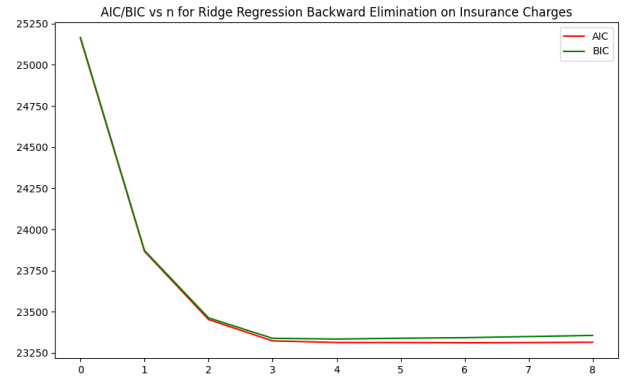
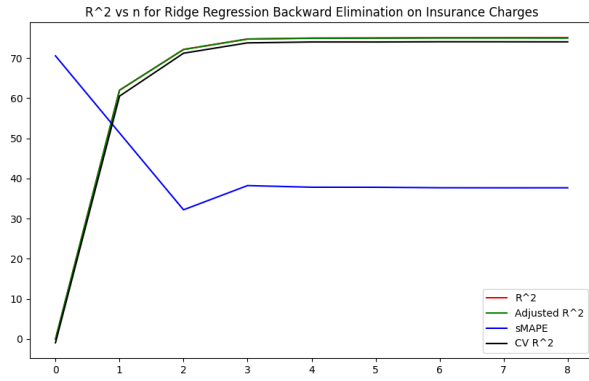


Figure 12: Insurance Charges Ridge Regression Backward Elimination Quality of Fit Metrics
Left: R^2 , Right: AIC/BIC

Stepwise Selection: Null, smoker yes, age, bmi, children, region southeast, region southwest

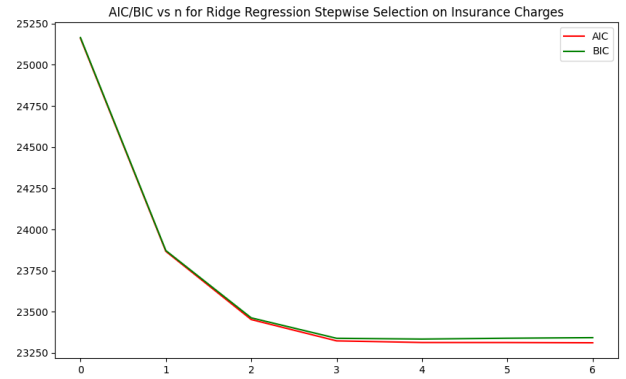
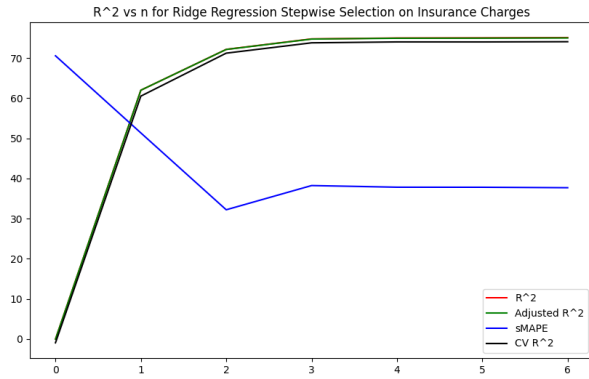


Figure 13: Insurance Charges Ridge Regression Stepwise Selection Quality of Fit Metrics
Left: R^2 , Right: AIC/BIC

3.4 Lasso Regression

3.4.1 Quality of Fit

Insurance Charges Lasso alpha Used: 0.0008

Insurance Charges Lasso Selected Features: age, bmi, children, sex male, smoker yes, region northwest, region southeast, region southwest

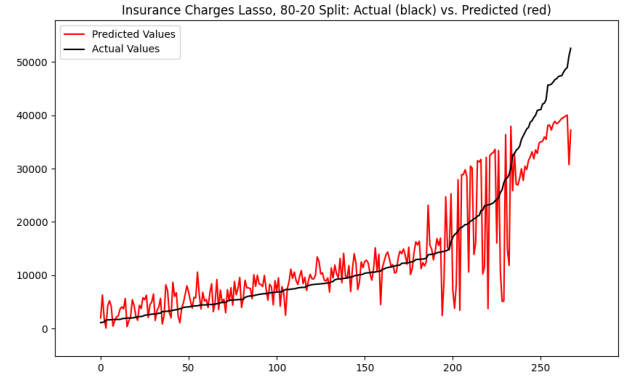
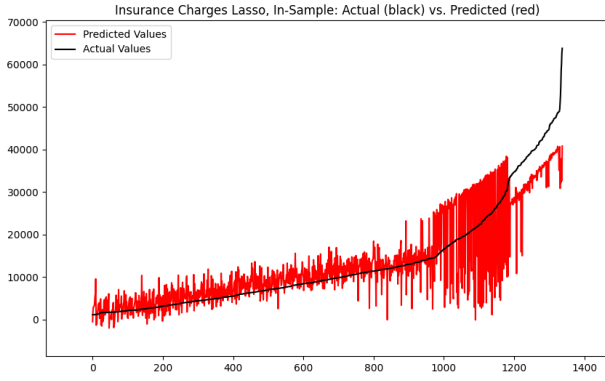


Figure 14: Insurance Charges Lasso Regression Plots
 Left: In-Sample, Right: Out-of-Sample
 black: actual y-values, red: predicted y-values

Table 5: Insurance Charges Lasso Regression

Metric	In-Sample	80-20 Split
rSq	0.7509	0.7997
rSqBar	0.7496	0.7943
sst	1.961e+11	4.265e+10
sse	4.884e+10	8.541e+09
sde	6060	5732
mse0	3.65e+07	3.187e+07
rmse	6042	5645
mae	4170	3936
smape	37.72	35.01
m	1338	268
dfr	8	8
df	1330	260
fStat	501.2	129.8
aic	2.331e+04	4646
bic	2.336e+04	4675

Table 6: Insurance Charges Lasso Regression CV

Metric	num folds	min	max	mean	stdev
rSq	5	0.6528	0.7997	0.7401	0.05184
rSqBar	5	0.6434	0.7943	0.7331	0.05324
sst	5	2.986e+10	4.265e+10	3.902e+10	4.668e+09
sse	5	8.541e+09	1.192e+10	9.938e+09	1.192e+09
sde	5	5732	6785	6176	371
mse0	5	3.187e+07	4.465e+07	3.714e+07	4.502e+06
rmse	5	5645	6682	6083	365.3
mae	5	3936	4560	4206	212.2
smape	5	35.01	41.12	37.94	2.044
m	5	267	268	267.6	0.4899
dfr	5	8	8	8	0
df	5	259	260	259.6	0.4899
fStat	5	61.09	129.8	97.17	24.09
aic	5	4646	4719	4678	26.69
bic	5	4675	4748	4707	26.68

3.4.2 Feature Selection

Forward Selection: Null, smoker yes, age, bmi, children, region southeast, region southwest, region northwest, sex male

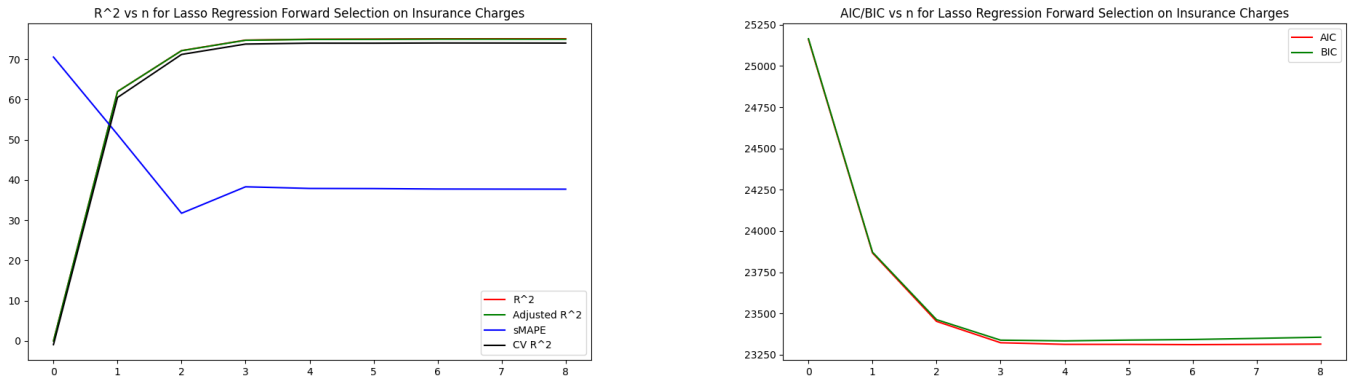


Figure 15: Insurance Charges Lasso Regression Forward Selection Quality of Fit Metrics
Left: R^2 , Right: AIC/BIC

Backward Elimination Reversed: Null, smoker yes, age, bmi, children, region southeast, region southwest, sex male, region northwest

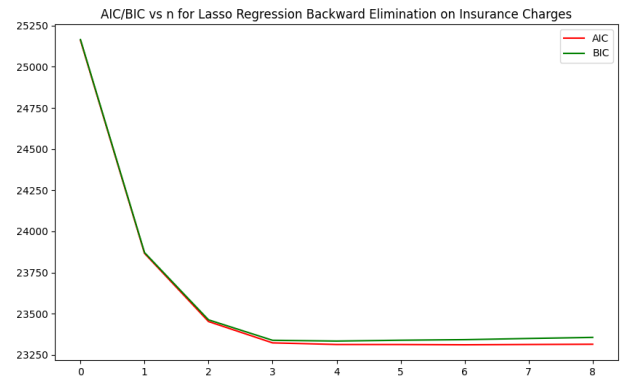
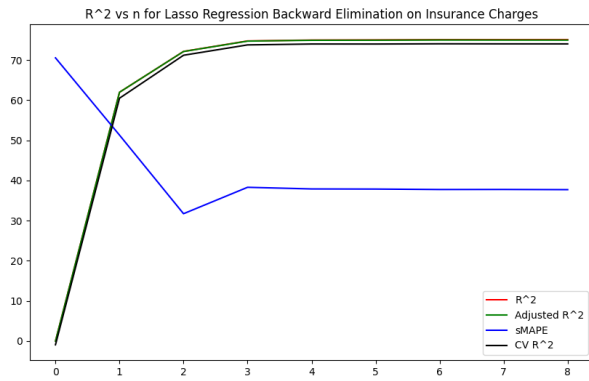


Figure 16: Insurance Charges Lasso Regression Backward Elimination Quality of Fit Metrics
Left: R^2 , Right: AIC/BIC

Stepwise Selection: Null, smoker yes, age, bmi, children, region southeast, region southwest

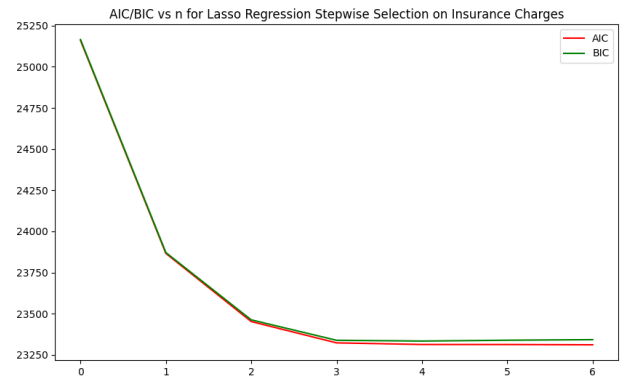
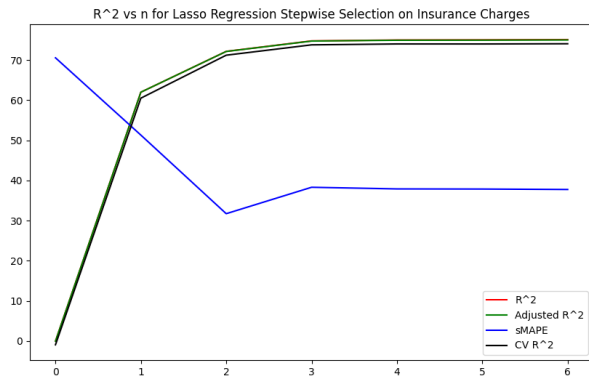


Figure 17: Insurance Charges Lasso Regression Stepwise Selection Quality of Fit Metrics
Left: R^2 , Right: AIC/BIC

3.5 Sqrt Transformation

3.5.1 Quality of Fit

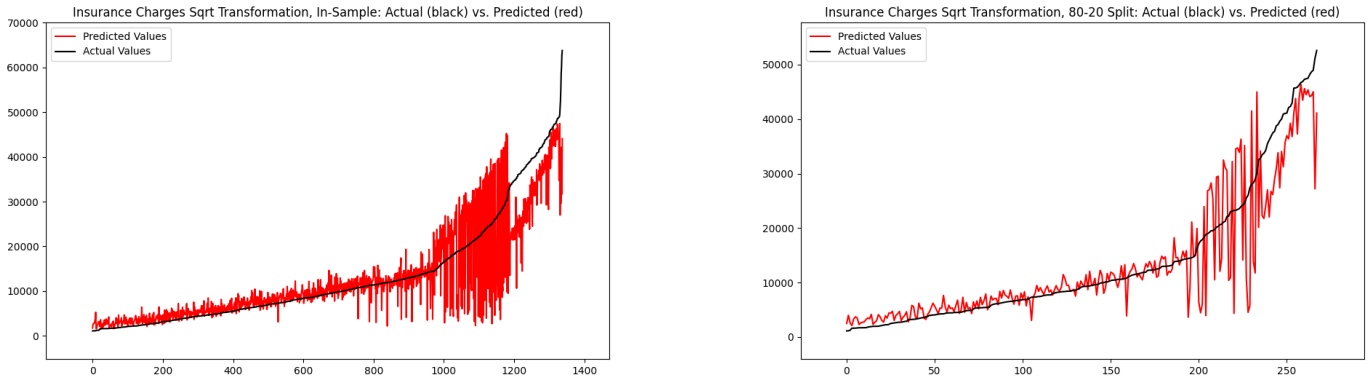


Figure 18: Insurance Charges Sqrt Transformation Plots
Left: In-Sample, Right: Out-of-Sample
black: actual y-values, red: predicted y-values

Table 7: Insurance Charges Sqrt Transformation

Metric	In-Sample	80-20 Split
rSq	0.7527	0.8077
rSqBar	0.7512	0.8017
sst	1.961e+11	4.265e+10
sse	4.85e+10	8.202e+09
sde	6041	5627
mse0	3.625e+07	3.06e+07
rmse	6020	5532
mae	3614	3316
smape	27.69	26.47
m	1338	268
dfr	9	9
df	1329	259
fStat	449.3	120.9
aic	2.331e+04	4637
bic	2.335e+04	4670

Table 8: Insurance Charges Sqrt Transformation CV

Metric	num folds	min	max	mean	stdev
rSq	5	0.6982	0.8077	0.7447	0.04001
rSqBar	5	0.6888	0.8017	0.7368	0.04125
sst	5	2.986e+10	4.265e+10	3.902e+10	4.668e+09
sse	5	8.202e+09	1.266e+10	9.869e+09	1.547e+09
sde	5	5627	7005	6160	476.5
mse0	5	3.06e+07	4.742e+07	3.689e+07	5.843e+06
rmse	5	5532	6886	6055	468.2
mae	5	3316	4146	3646	299.2
smape	5	26.47	29.49	27.88	1.068
m	5	267	268	267.6	0.4899
dfr	5	9	9	9	0
df	5	258	259	258.6	0.4899
fStat	5	66.31	120.9	86.85	19.67
aic	5	4637	4737	4677	33.57
bic	5	4670	4769	4710	33.55

3.5.2 Feature Selection

Forward Selection: Null, smoker yes, bmi, age, region southeast, region southwest, region northwest, sex male, children

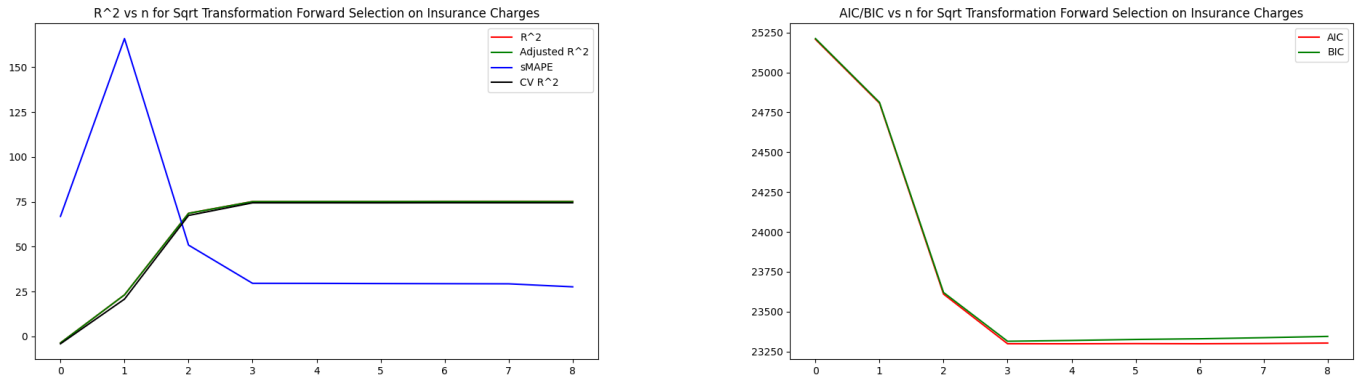


Figure 19: Insurance Charges Sqrt Transformation Forward Selection Quality of Fit Metrics
Left: R^2 , Right: AIC/BIC

Backward Elimination Reversed: Null, smoker yes, bmi, age, region southeast, region southwest, region northwest, sex male, children

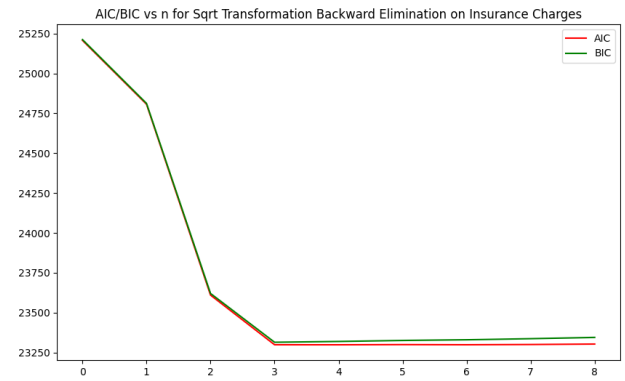
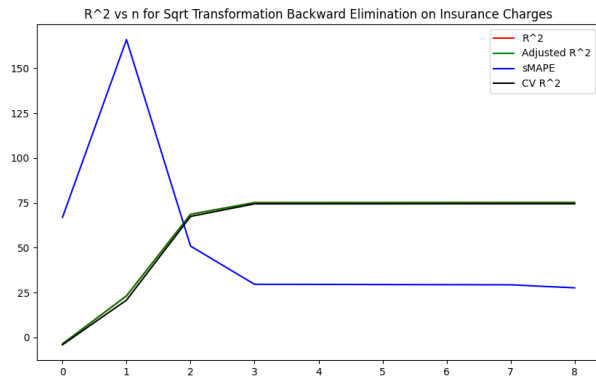


Figure 20: Insurance Charges Sqrt Transformation Backward Elimination Quality of Fit Metrics
Left: R^2 , Right: AIC/BIC

Stepwise Selection: Null, smoker yes, bmi, age, region southeast

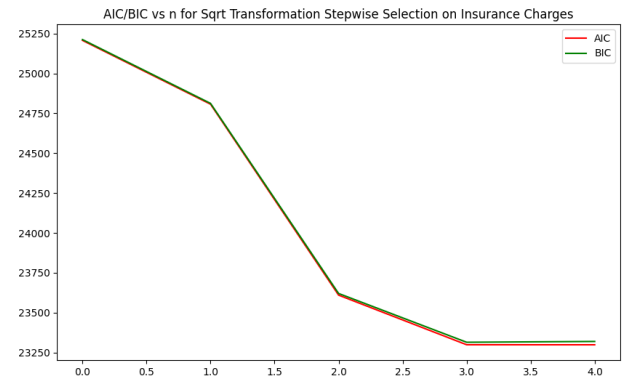
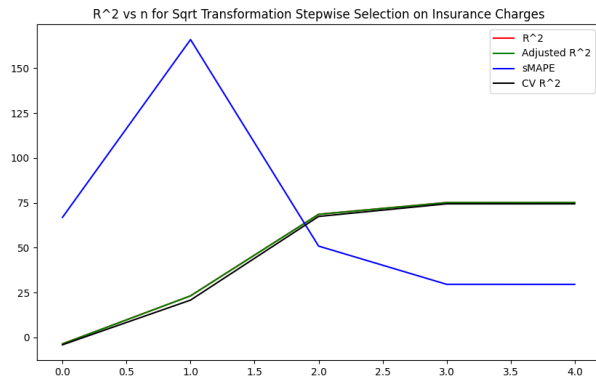


Figure 21: Insurance Charges Sqrt Transformation Stepwise Selection Quality of Fit Metrics
Left: R^2 , Right: AIC/BIC

3.6 Log1p Transformation

3.6.1 Quality of Fit

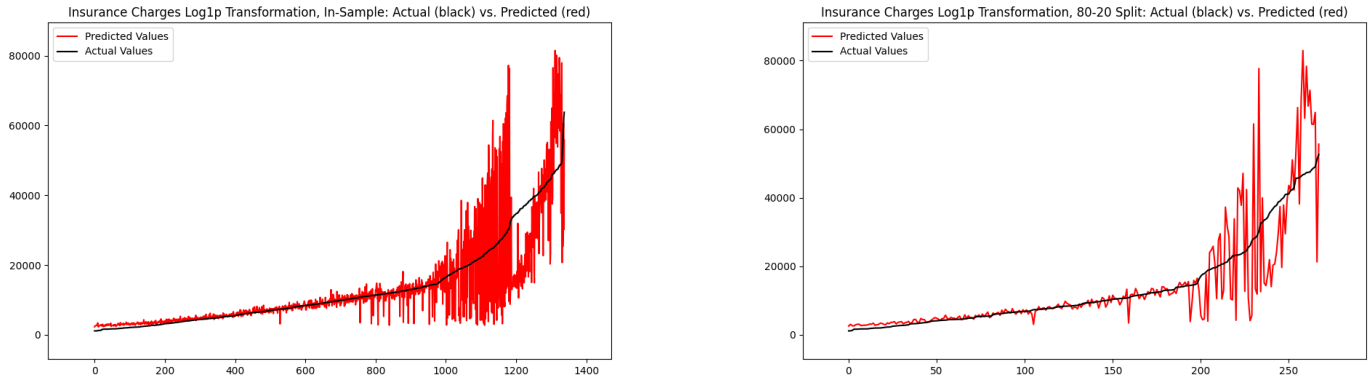


Figure 22: Insurance Charges Log1p Transformation Plots
Left: In-Sample, Right: Out-of-Sample
black: actual y-values, red: predicted y-values

Table 9: Insurance Charges Log1p Transformation

Metric	In-Sample	80-20 Split
rSq	0.5228	0.5468
rSqBar	0.5199	0.5328
sst	1.961e+11	4.265e+10
sse	9.357e+10	1.933e+10
sde	8391	8638
mse0	6.993e+07	7.211e+07
rmse	8363	8492
mae	4220	4213
smape	26.29	25.08
m	1338	268
dfr	9	9
df	1329	259
fStat	161.8	34.73
aic	2.419e+04	4867
bic	2.423e+04	4899

Table 10: Insurance Charges Log1p Transformation CV

Metric	num folds	min	max	mean	stdev
rSq	5	0.4531	0.5468	0.5104	0.03686
rSqBar	5	0.4361	0.5328	0.4953	0.03803
sst	5	2.986e+10	4.265e+10	3.902e+10	4.668e+09
sse	5	1.395e+10	2.294e+10	1.915e+10	3.002e+09
sde	5	7339	9430	8578	702.2
mse0	5	5.205e+07	8.592e+07	7.159e+07	1.132e+07
rmse	5	7215	9269	8433	690.1
mae	5	3509	4865	4275	448.8
smape	5	25.08	28.42	26.53	1.153
m	5	267	268	267.6	0.4899
dfr	5	9	9	9	0
df	5	258	259	258.6	0.4899
fStat	5	23.75	34.73	30.29	4.334
aic	5	4780	4896	4854	39.38
bic	5	4812	4928	4887	39.37

3.6.2 Feature Selection

Forward Selection: Null, smoker yes, sex male, bmi, age, region northwest, region southwest, children, region southeast

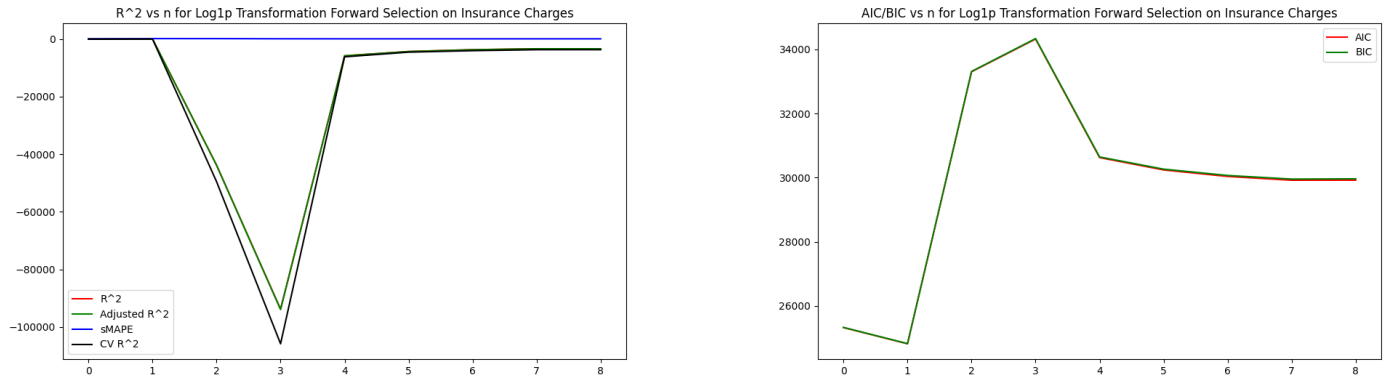


Figure 23: Insurance Charges Log1p Transformation Forward Selection Quality of Fit Metrics
Left: R^2 , Right: AIC/BIC

Backward Elimination Reversed: Null, age, bmi, region northwest, region southwest, sex male, children, region southeast, smoker yes

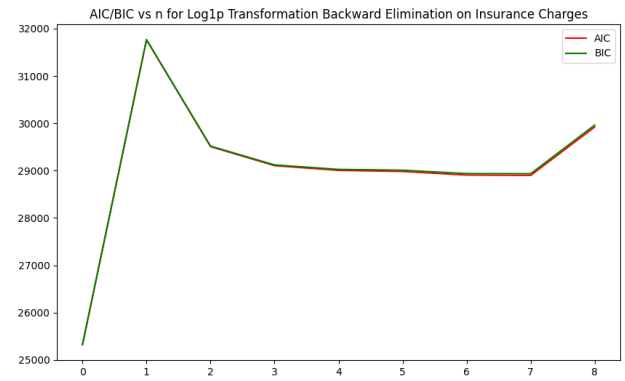
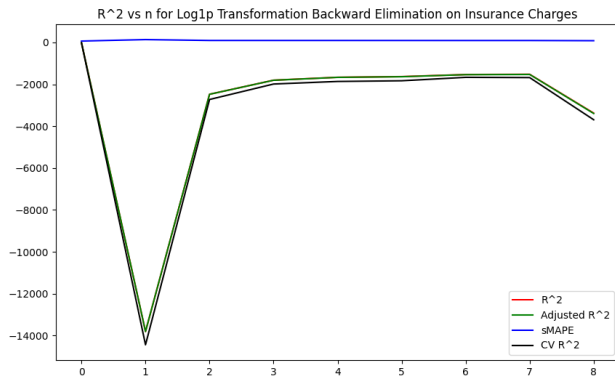


Figure 24: Insurance Charges Log1p Transformation Backward Elimination Quality of Fit Metrics
Left: R^2 , Right: AIC/BIC

Stepwise Selection: Null, smoker yes

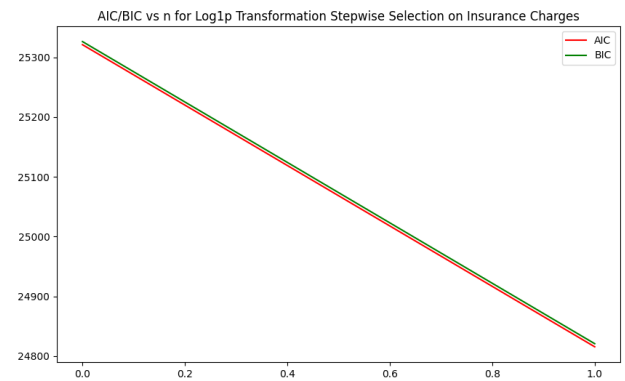
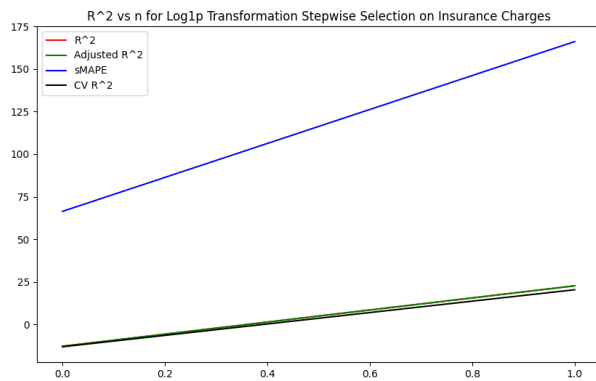


Figure 25: Insurance Charges Log1p Transformation Stepwise Selection Quality of Fit Metrics
Left: R^2 , Right: AIC/BIC

3.7 Box-Cox Transformation

3.7.1 Quality of Fit

Insurance Charges Box-Cox lambda Used: 0.5

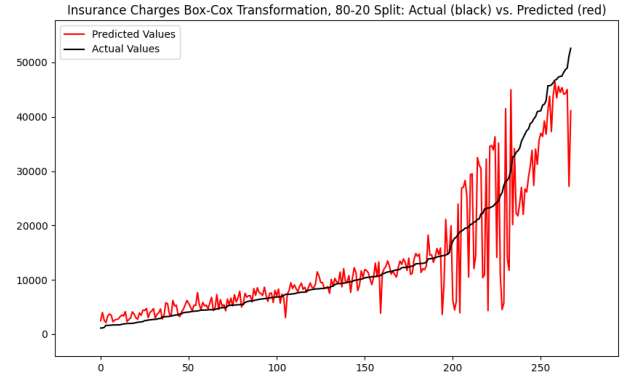
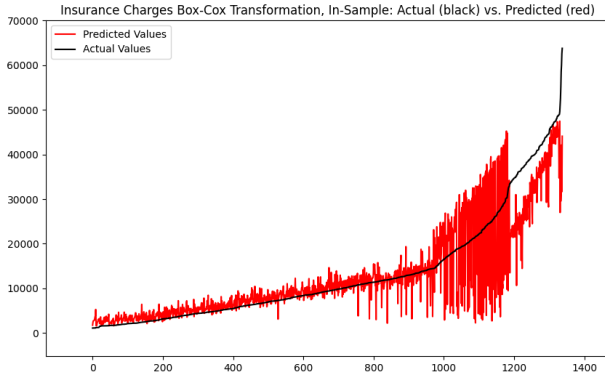


Figure 26: Insurance Charges Box-Cox Transformation Plots
 Left: In-Sample, Right: Out-of-Sample
 black: actual y-values, red: predicted y-values

Table 11: Insurance Charges Box-Cox Transformation

Metric	In-Sample	80-20 Split
rSq	0.7527	0.8077
rSqBar	0.7512	0.8017
sst	1.961e+11	4.265e+10
sse	4.85e+10	8.202e+09
sde	6041	5627
mse0	3.625e+07	3.06e+07
rmse	6020	5532
mae	3614	3316
smape	27.69	26.47
m	1338	268
dfr	9	9
df	1329	259
fStat	449.3	120.9
aic	2.331e+04	4637
bic	2.335e+04	4670

Table 12: Insurance Charges Box-Cox Transformation CV

Metric	num folds	min	max	mean	stdev
rSq	5	0.6982	0.8077	0.7447	0.04001
rSqBar	5	0.6888	0.8017	0.7368	0.04125
sst	5	2.986e+10	4.265e+10	3.902e+10	4.668e+09
sse	5	8.202e+09	1.266e+10	9.869e+09	1.547e+09
sde	5	5627	7005	6160	476.5
mse0	5	3.06e+07	4.742e+07	3.689e+07	5.843e+06
rmse	5	5532	6886	6055	468.2
mae	5	3316	4146	3646	299.2
smape	5	26.47	29.49	27.88	1.068
m	5	267	268	267.6	0.4899
dfr	5	9	9	9	0
df	5	258	259	258.6	0.4899
fStat	5	66.31	120.9	86.85	19.67
aic	5	4637	4737	4677	33.57
bic	5	4670	4769	4710	33.55

3.7.2 Feature Selection

Forward Selection: Null, smoker yes, bmi, age, region southeast, region southwest, region northwest, sex male, children

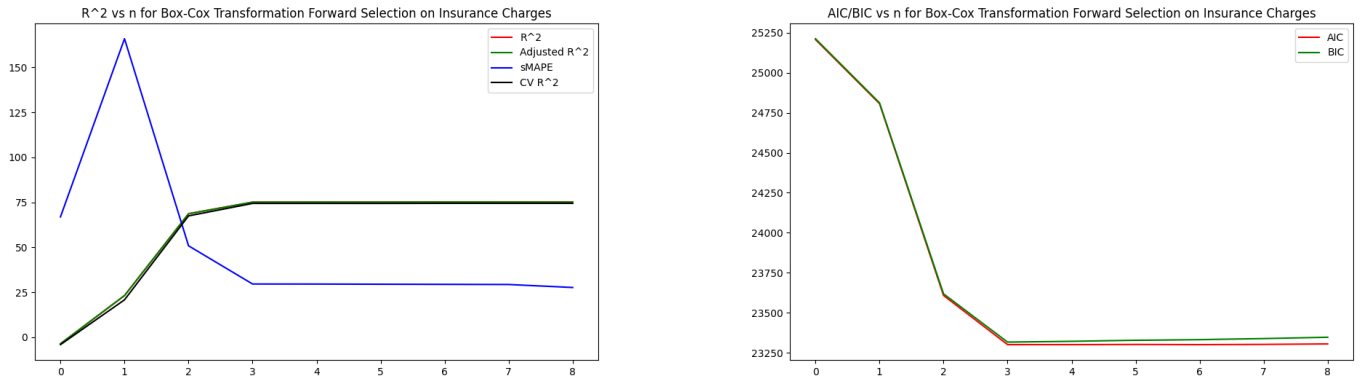


Figure 27: Insurance Charges Box-Cox Transformation Forward Selection Quality of Fit Metrics
Left: R^2 , Right: AIC/BIC

Backward Elimination Reversed: Null, smoker yes, bmi, age, region southeast, region southwest, region northwest, sex male, children

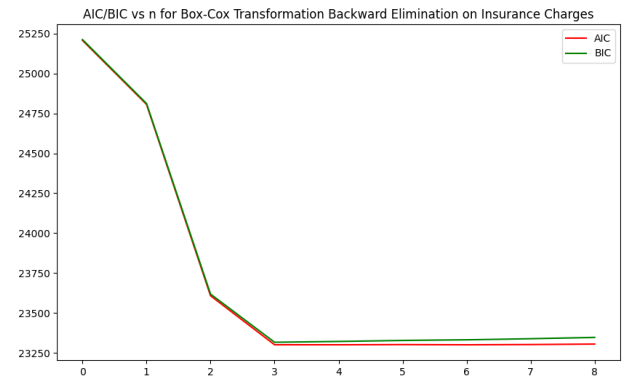
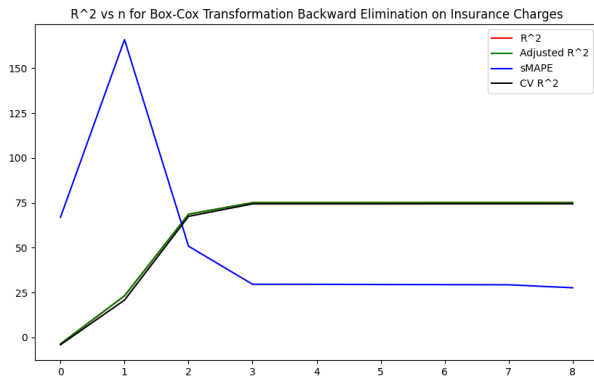


Figure 28: Insurance Charges Box-Cox Transformation Backward Elimination Quality of Fit Metrics
Left: R^2 , Right: AIC/BIC

Stepwise Selection: Null, smoker yes, bmi, age, region southeast

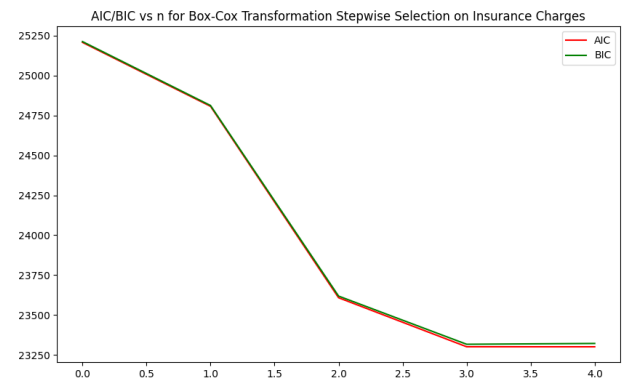
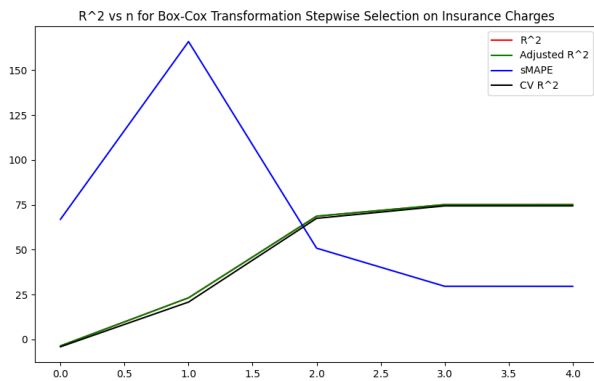


Figure 29: Insurance Charges Box-Cox Transformation Stepwise Selection Quality of Fit Metrics
Left: R^2 , Right: AIC/BIC

3.8 Order 2 Regression

3.8.1 Quality of Fit

Insurance Charges Order 2 Polynomial alpha Used: 0.0008500000000000002

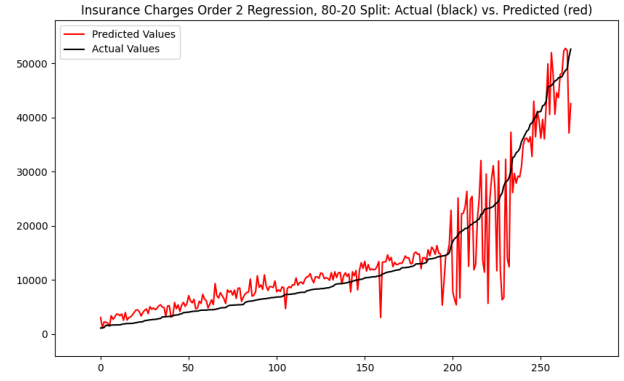
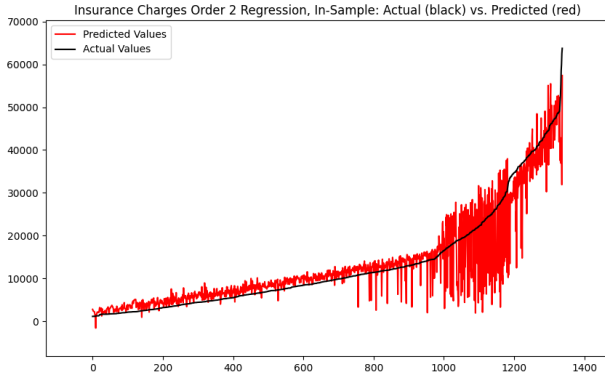


Figure 30: Insurance Charges Order 2 Regression Plots
 Left: In-Sample, Right: Out-of-Sample
 black: actual y-values, red: predicted y-values

Table 13: Insurance Charges Order 2 Polynomial Regression

Metric	In-Sample	80-20 Split
rSq	0.8477	0.88
rSqBar	0.8406	0.8452
sst	1.961e+11	4.265e+10
sse	2.986e+10	5.117e+09
sde	4835	4972
mse0	2.232e+07	1.909e+07
rmse	4724	4370
mae	2833	2957
smape	26.43	28.54
m	1338	268
dfr	61	61
df	1277	207
fStat	116.5	24.89
aic	2.276e+04	4615
bic	2.308e+04	4834

Table 14: Insurance Charges Order 2 Polynomial Regression CV

Metric	num folds	min	max	mean	stdev
rSq	5	0.7851	0.88	0.836	0.0399
rSqBar	5	0.7225	0.8452	0.7884	0.05152
sst	5	2.986e+10	4.265e+10	3.902e+10	4.668e+09
sse	5	5.117e+09	9.015e+09	6.302e+09	1.412e+09
sde	5	4972	6615	5492	590.4
mse0	5	1.909e+07	3.377e+07	2.356e+07	5.312e+06
rmse	5	4370	5811	4826	517.8
mae	5	2616	3289	2913	220.3
smape	5	25.79	28.54	26.99	0.9621
m	5	267	268	267.6	0.4899
dfr	5	61	61	61	0
df	5	206	207	206.6	0.4899
fStat	5	12.34	24.89	18.49	5.052
aic	5	4615	4750	4658	49.31
bic	5	4834	4969	4877	49.25

3.8.2 Feature Selection

Forward Selection: Null, bmi x smoker yes, age x age, smoker no, children x smoker no, bmi x region northeast, children x region northwest, age x region northeast, children x sex female, smoker yes x region southwest, children x region southwest, smoker yes x region southeast, sex male x region northeast, age x region northwest, bmi x region northwest, bmi x bmi, bmi, children, children x children, region northwest, sex male x region southwest, age x children, bmi x smoker no, children x region northeast, age x region southwest, bmi x region southwest, smoker no x region southeast, age x sex female, sex female, sex male x region northwest, smoker no x region northwest, smoker yes, smoker no x region northeast, age x smoker yes, age, age x bmi, bmi x sex male, bmi x children, sex male x smoker yes, sex female x smoker yes, bmi x sex female, sex female x smoker no, sex male x smoker no, smoker yes x region northwest, smoker yes x region northeast, children x sex male, age x smoker no, smoker no x region southwest, children x region southeast, sex female x region northeast, sex male, region southwest, sex female x region southwest, sex male x region southeast, region northeast, sex female x region northwest, sex female x region southeast, region southeast, children x smoker yes, age x sex male, age x region southeast, bmi x region southeast

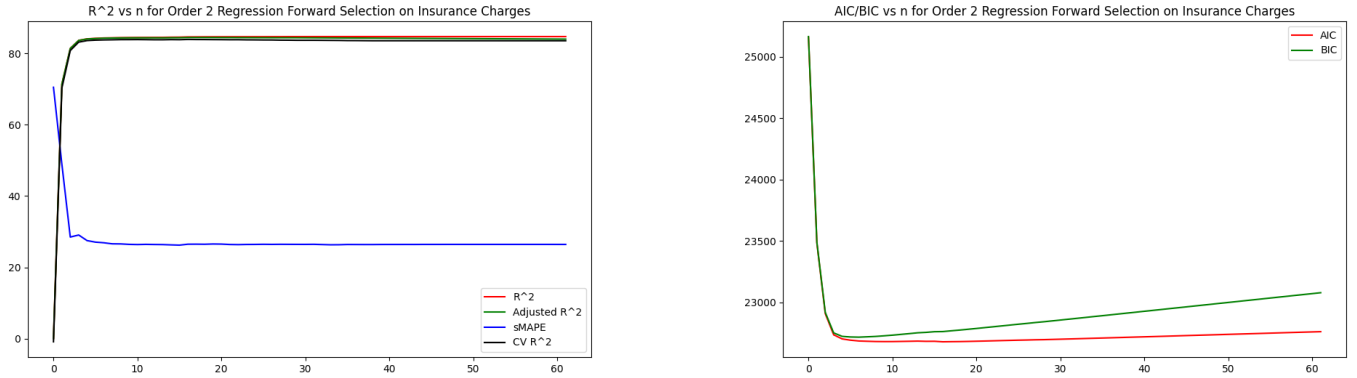


Figure 31: Insurance Charges Order 2 Regression Forward Selection Quality of Fit Metrics
Left: R^2 , Right: AIC/BIC

Backward Elimination Reversed: Null, bmi x smoker yes, age x age, smoker yes, children x smoker no, bmi x region northeast, bmi x bmi, bmi, children x region northwest, children x sex female, age x region northeast, age x region northwest, bmi x region northwest, smoker yes x region southwest, smoker no x region southeast, children x region northeast, sex female x smoker no, sex male x smoker yes, age x sex female, children x children, children x region southeast, bmi x region southwest, sex female x region northeast, smoker yes x region southeast, children, bmi x smoker no, age x children, sex male x region southwest, sex male x region northwest, age, age x bmi, smoker yes x region northwest, smoker no, smoker no x region southwest, smoker no

x region northeast, age x region southeast, age x smoker yes, sex female x smoker yes, bmi x children, bmi x sex male, smoker yes x region northeast, bmi x sex female, sex male x smoker no, age x smoker no, children x sex male, sex male, smoker no x region northwest, sex female, region northwest, region southwest, sex female x region southwest, sex male x region southeast, sex female x region northwest, sex male x region northeast, region northeast, sex female x region southeast, region southeast, children x smoker yes, children x region southwest, age x sex male, age x region southwest, bmi x region southeast

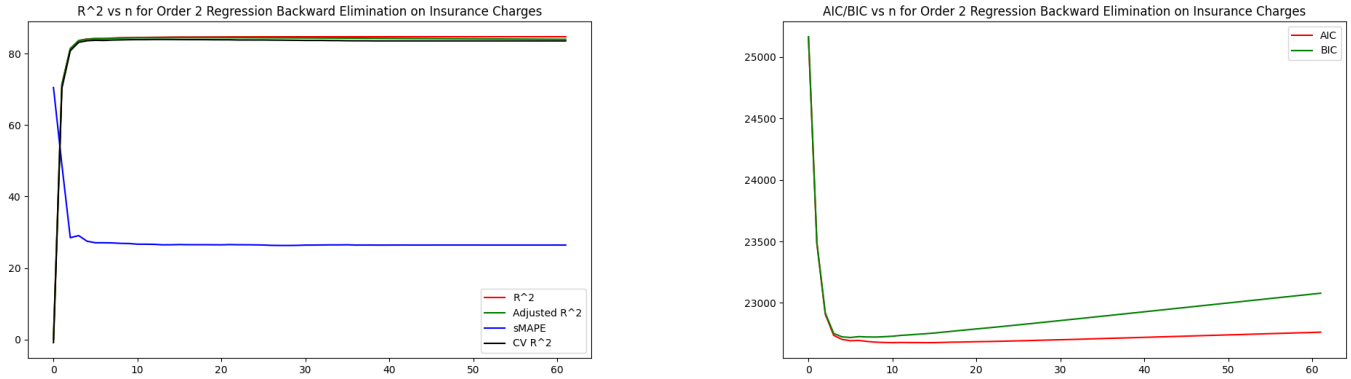


Figure 32: Insurance Charges Order 2 Regression Backward Elimination Quality of Fit Metrics
Left: R^2 , Right: AIC/BIC

Stepwise Selection: Null, bmi x smoker yes, age x age, smoker no, children x smoker no, bmi x region northeast, children x region northwest, age x region northeast, children x sex female, smoker yes x region southwest, children x region southwest

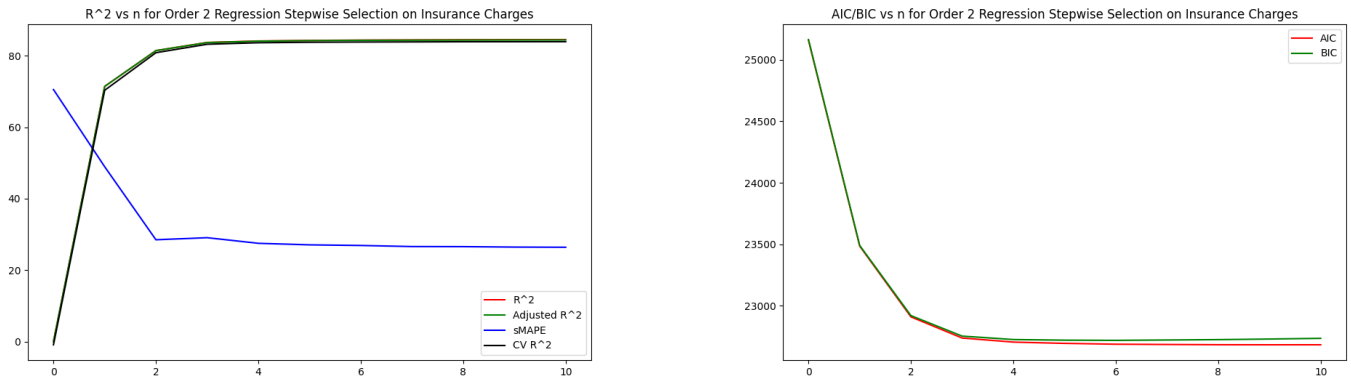


Figure 33: Insurance Charges Order 2 Regression Stepwise Selection Quality of Fit Metrics
Left: R^2 , Right: AIC/BIC

3.9 Two Layer Neural Network

3.9.1 Quality of Fit

Insurance Charges NN 2L output activation: ELU, lr: 5e-03

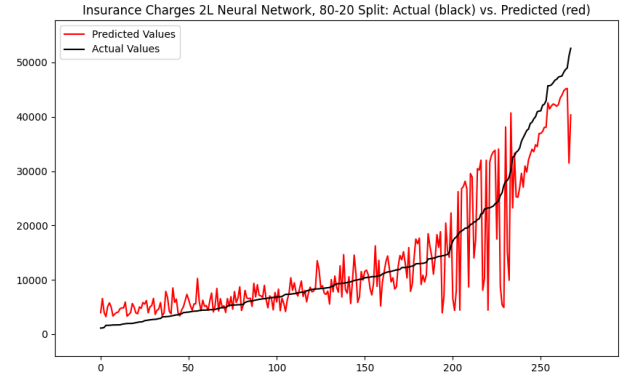
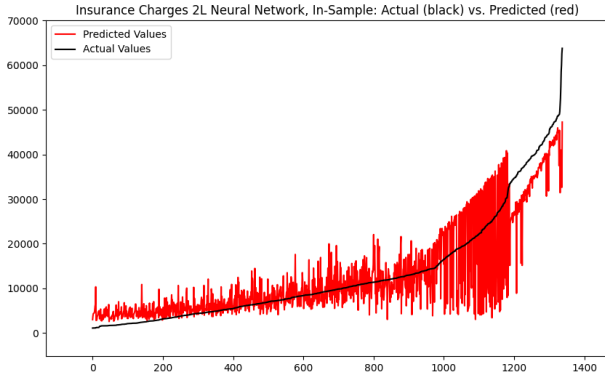


Figure 34: Insurance Charges 2 Layer Neural Network Plots
 Left: In-Sample, Right: Out-of-Sample
 black: actual y-values, red: predicted y-values

Table 15: Insurance Charges 2L Neural Network

Metric	In-Sample	80-20 Split
rSq	0.7703	0.8211
rSqBar	0.7691	0.8163
sst	1.961e+11	4.265e+10
sse	4.504e+10	7.631e+09
sde	5819	5418
mse0	3.366e+07	2.847e+07
rmse	5802	5336
mae	3996	3668
smape	35.78	33.65
m	1338	268
dfr	8	8
df	1330	260
fStat	557.5	149.1
aic	2.321e+04	4616
bic	2.325e+04	4645

Table 16: Insurance Charges 2L Neural Network CV

Metric	num folds	min	max	mean	stdev
rSq	5	0.6721	0.8208	0.7589	0.05256
rSqBar	5	0.6632	0.816	0.7524	0.05397
sst	5	2.986e+10	4.265e+10	3.902e+10	4.668e+09
sse	5	7.643e+09	1.13e+10	9.209e+09	1.279e+09
sde	5	5422	6604	5942	413.2
mse0	5	2.852e+07	4.231e+07	3.442e+07	4.822e+06
rmse	5	5340	6505	5852	406.8
mae	5	3668	4449	4045	256.5
smape	5	33.62	39.57	36.06	2.255
m	5	267	268	267.6	0.4899
dfr	5	8	8	8	0
df	5	259	260	259.6	0.4899
fStat	5	66.6	148.9	108.3	28.85
aic	5	4616	4705	4657	32.09
bic	5	4645	4733	4686	32.08

3.9.2 Feature Selection

Forward Selection: Null, smoker yes, age, bmi, children, region southeast, region southwest, sex male, region northwest

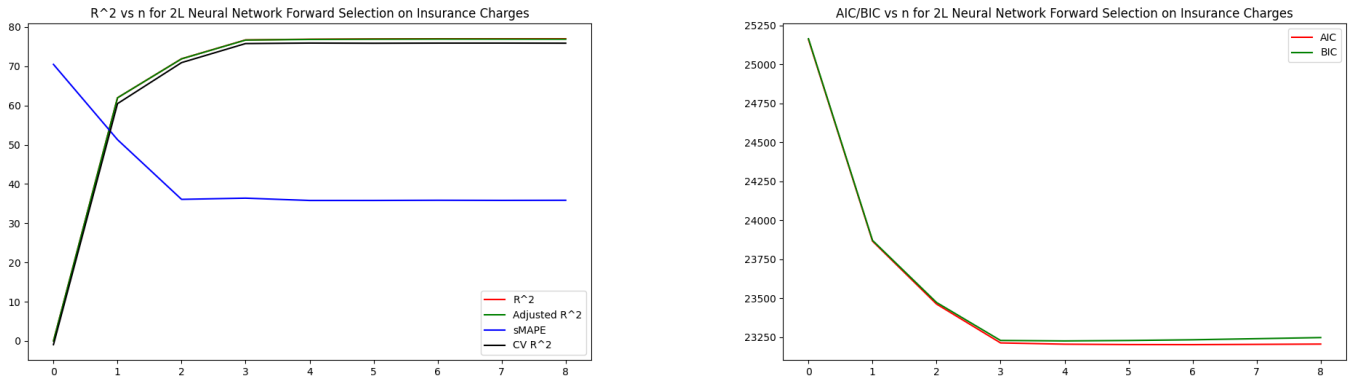


Figure 35: Insurance Charges 2 Layer Neural Network Forward Selection Quality of Fit Metrics
Left: R^2 , Right: AIC/BIC

Backward Elimination Reversed: Null, smoker yes, age, bmi, children, region southeast, region southwest, sex male, region northwest

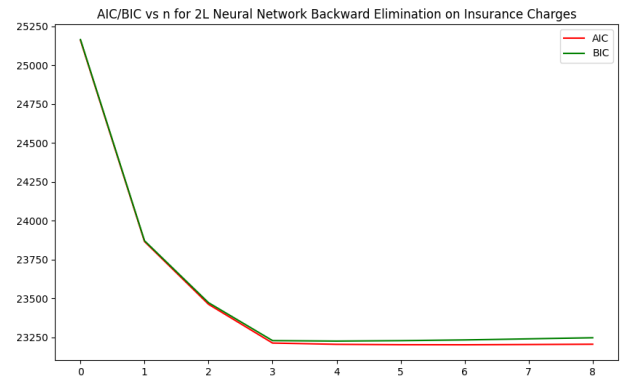
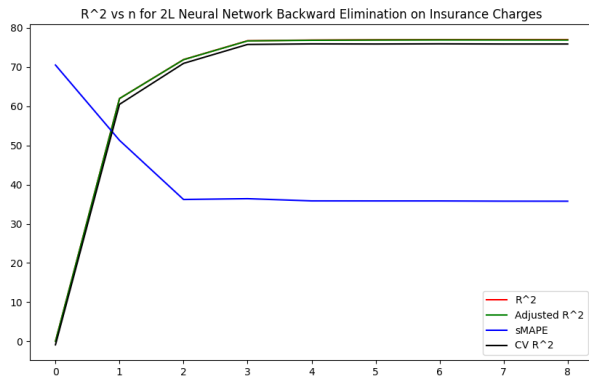


Figure 36: Insurance Charges 2 Layer Neural Network Backward Elimination Quality of Fit Metrics
Left: R^2 , Right: AIC/BIC

Stepwise Selection: Null, smoker yes, age, bmi, children, region southeast, region southwest

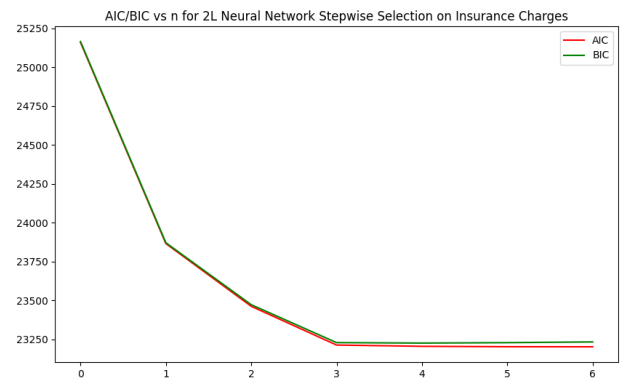
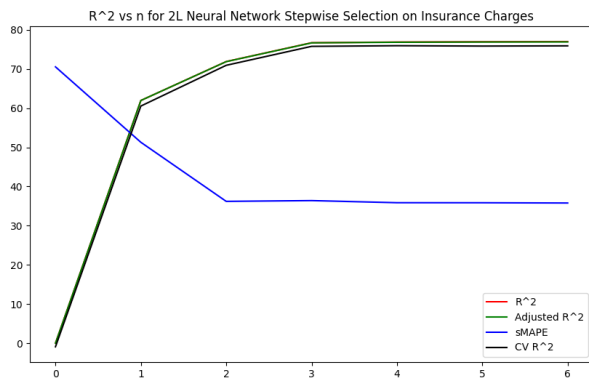


Figure 37: Insurance Charges 2 Layer Neural Network Stepwise Selection Quality of Fit Metrics
Left: R^2 , Right: AIC/BIC

3.10 Three Layer Neural Network

3.10.1 Quality of Fit

Insurance Charges NN 3L hidden activation: ELU, output activation: Identity, hidden 1: 8, lr: 5e-02

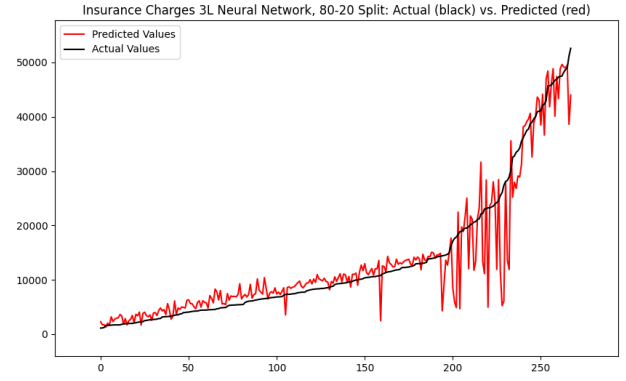
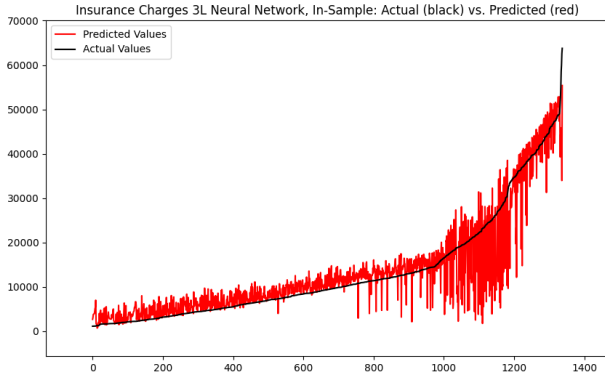


Figure 38: Insurance Charges 3 Layer Neural Network Plots
 Left: In-Sample, Right: Out-of-Sample
 black: actual y-values, red: predicted y-values

Table 17: Insurance Charges 3L Neural Network

Metric	In-Sample	80-20 Split
rSq	0.8535	0.8926
rSqBar	0.8527	0.8897
sst	1.961e+11	4.265e+10
sse	2.873e+10	4.58e+09
sde	4648	4197
mse0	2.147e+07	1.709e+07
rmse	4634	4134
mae	2814	2401
smape	27.01	23.25
m	1338	268
dfr	8	8
df	1330	260
fStat	968.4	270.2
aic	2.26e+04	4479
bic	2.265e+04	4508

Table 18: Insurance Charges 3L Neural Network CV

Metric	num folds	min	max	mean	stdev
rSq	5	0.7925	0.8834	0.8427	0.03985
rSqBar	5	0.7869	0.8803	0.8384	0.04093
sst	5	2.986e+10	4.265e+10	3.902e+10	4.668e+09
sse	5	4.971e+09	8.555e+09	6.03e+09	1.333e+09
sde	5	4372	5747	4793	508.9
mse0	5	1.855e+07	3.204e+07	2.254e+07	5.012e+06
rmse	5	4307	5660	4721	501.2
mae	5	2347	3363	2831	337.5
smape	5	24.27	34.1	28.2	3.861
m	5	267	268	267.6	0.4899
dfr	5	8	8	8	0
df	5	259	260	259.6	0.4899
fStat	5	124.1	246.3	186.8	51.41
aic	5	4501	4630	4541	49.82
bic	5	4530	4659	4570	49.81

3.10.2 Feature Selection

Forward Selection: Null, smoker yes, bmi, age, children, region southwest, region southeast, region northwest, sex male

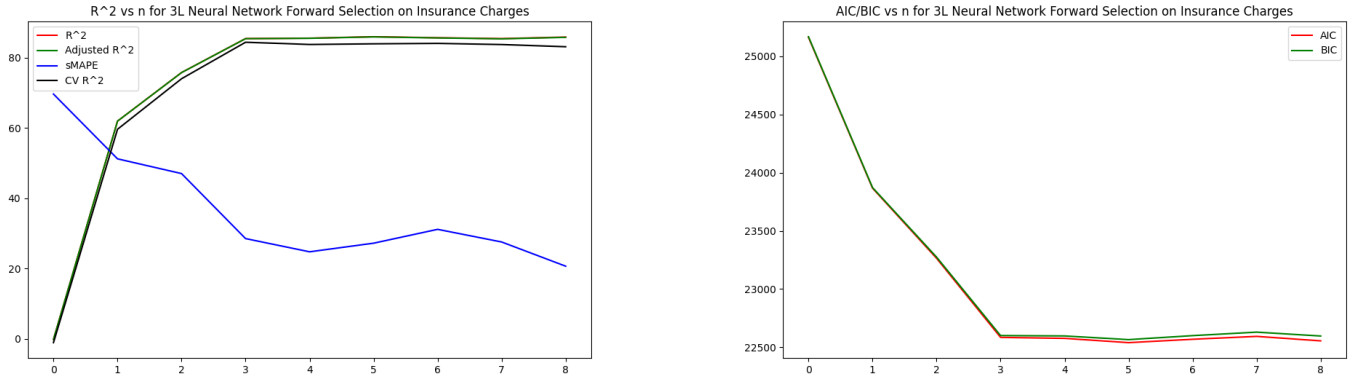


Figure 39: Insurance Charges 3 Layer Neural Network Forward Selection Quality of Fit Metrics
Left: R^2 , Right: AIC/BIC

Backward Elimination Reversed: Null, smoker yes, bmi, age, region northwest, children, region southwest, sex male, region southeast

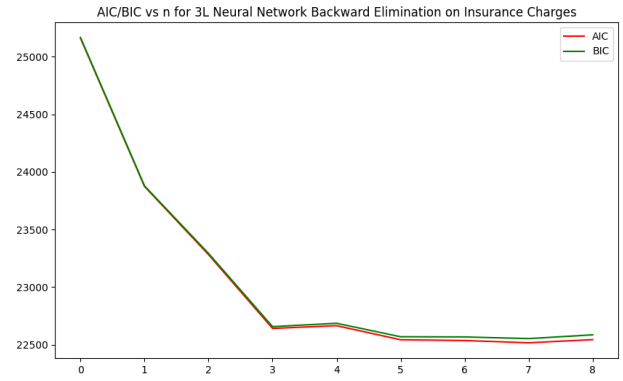
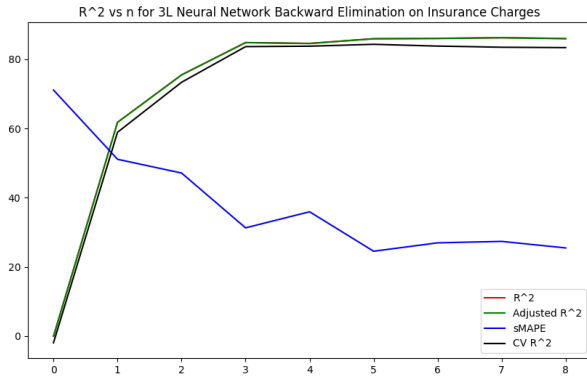


Figure 40: Insurance Charges 3 Layer Neural Network Backward Elimination Quality of Fit Metrics
Left: R^2 , Right: AIC/BIC

Stepwise Selection: Null, smoker yes, bmi, age, children

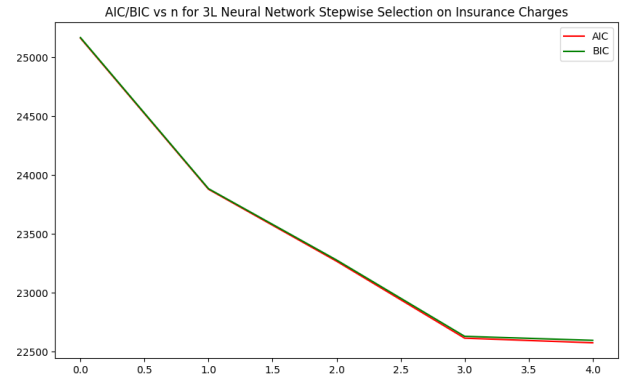
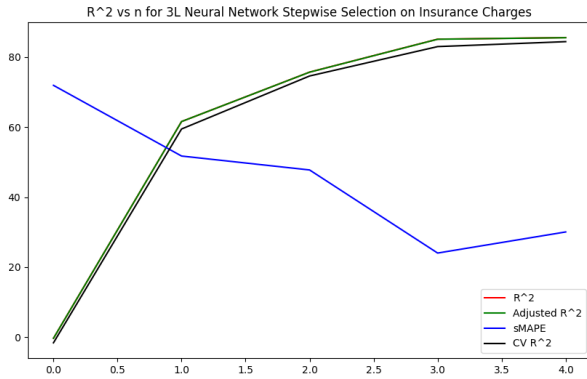


Figure 41: Insurance Charges 3 Layer Neural Network Stepwise Selection Quality of Fit Metrics
Left: R^2 , Right: AIC/BIC

3.11 Four Layer Neural Network

3.11.1 Quality of Fit

Insurance Charges NN 4L hidden activation 1: ELU, hidden activation 2: ELU, output activation: Identity, hidden 1: 256, hidden 2: 32, lr: 5e-03

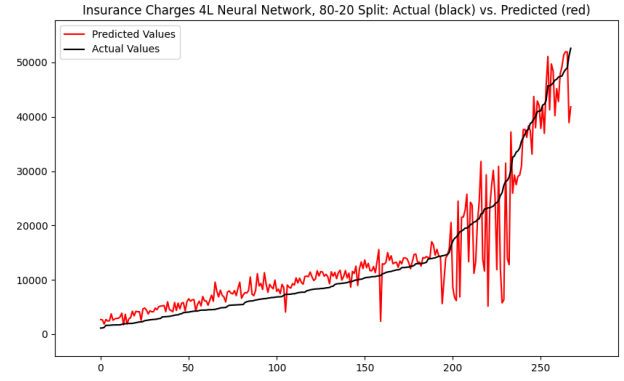
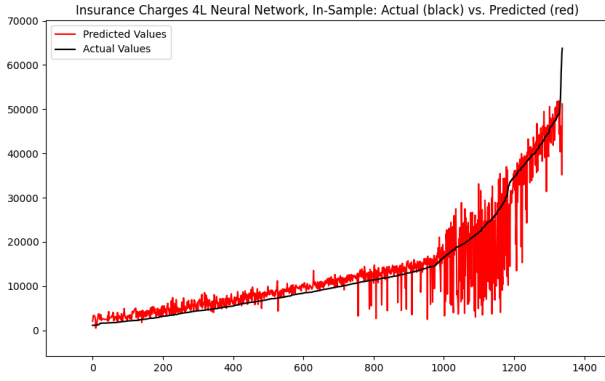


Figure 42: Insurance Charges 4 Layer Neural Network Plots
Left: In-Sample, Right: Out-of-Sample
black: actual y-values, red: predicted y-values

Table 19: Insurance Charges 4L Neural Network

Metric	In-Sample	80-20 Split
rSq	0.8674	0.884
rSqBar	0.8667	0.8809
sst	1.961e+11	4.265e+10
sse	2.6e+10	4.947e+09
sde	4421	4362
mse0	1.943e+07	1.846e+07
rmse	4408	4296
mae	2557	2900
smape	25.04	28.72
m	1338	268
dfr	8	8
df	1330	260
fStat	1088	247.7
aic	2.247e+04	4500
bic	2.251e+04	4529

Table 20: Insurance Charges 4L Neural Network CV

Metric	num folds	min	max	mean	stdev
rSq	5	0.7924	0.8866	0.842	0.0407
rSqBar	5	0.7868	0.8836	0.8377	0.0418
sst	5	2.986e+10	4.265e+10	3.902e+10	4.668e+09
sse	5	4.835e+09	8.653e+09	6.057e+09	1.381e+09
sde	5	4312	5780	4802	527.6
mse0	5	1.804e+07	3.241e+07	2.264e+07	5.194e+06
rmse	5	4248	5693	4730	519.6
mae	5	2476	3114	2786	228.3
smape	5	25.74	29.89	27.06	1.484
m	5	267	268	267.6	0.4899
dfr	5	8	8	8	0
df	5	259	260	259.6	0.4899
fStat	5	124	254.1	186.6	53.39
aic	5	4494	4633	4542	51.25
bic	5	4523	4662	4570	51.24

3.11.2 Feature Selection

Forward Selection: Null, smoker yes, bmi, age, children, region northwest, sex male, region southwest, region southeast

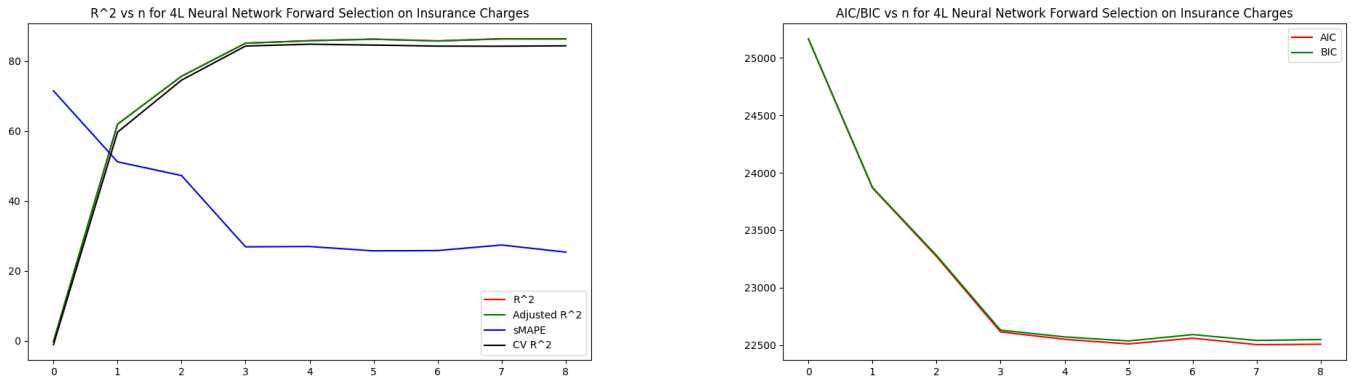


Figure 43: Insurance Charges 4 Layer Neural Network Forward Selection Quality of Fit Metrics
Left: R^2 , Right: AIC/BIC

Backward Elimination Reversed: Null, smoker yes, bmi, age, children, region southeast, region southwest, sex male, region northwest

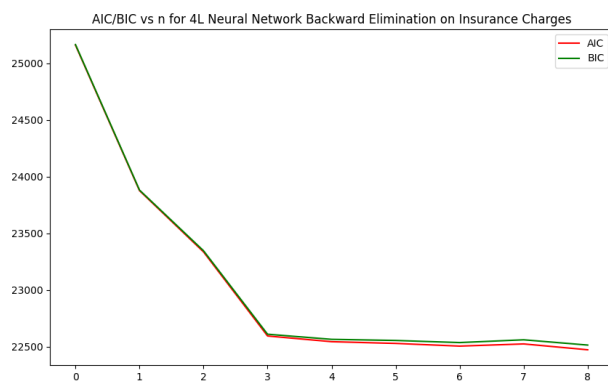
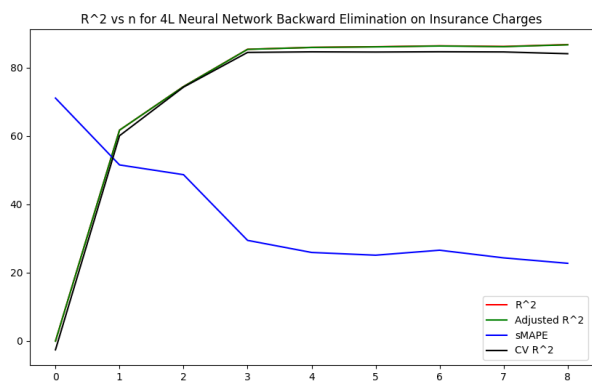


Figure 44: Insurance Charges 4 Layer Neural Network Backward Elimination Quality of Fit Metrics
Left: R^2 , Right: AIC/BIC

Stepwise Selection: Null, smoker yes, bmi, age, children, region southeast, region southwest, sex male

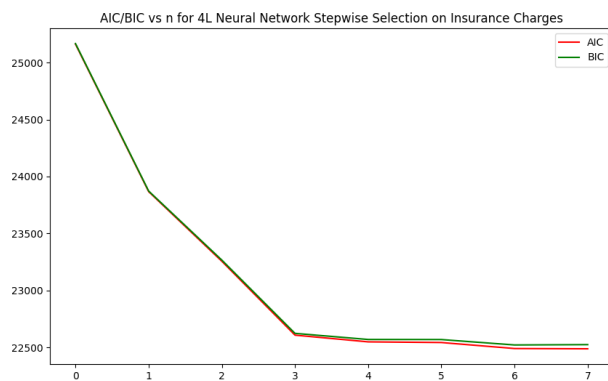
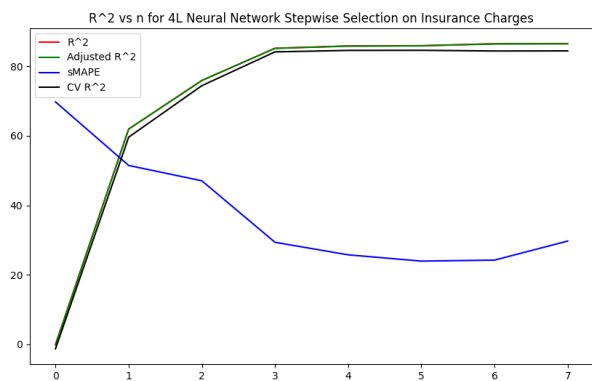


Figure 45: Insurance Charges 4 Layer Neural Network Stepwise Selection Quality of Fit Metrics
Left: R^2 , Right: AIC/BIC

3.12 Random Forest

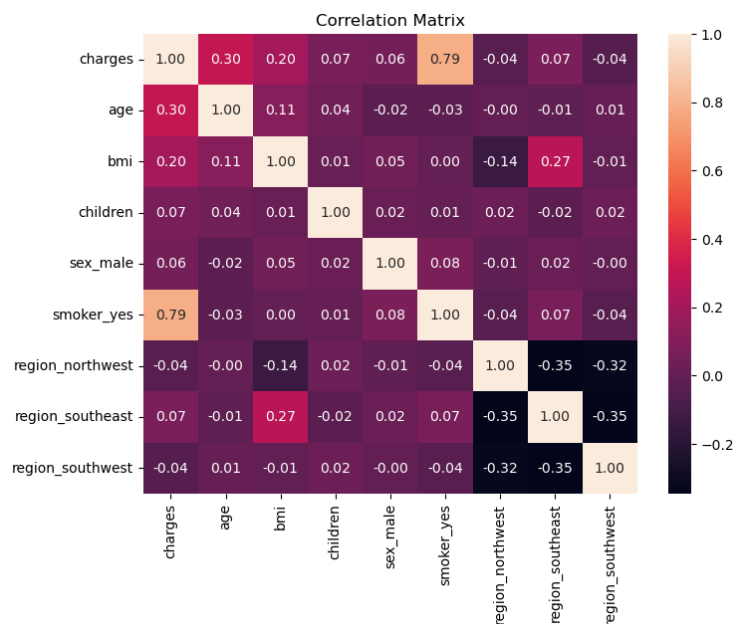


Figure 46: Insurance Charges Correlation Matrix

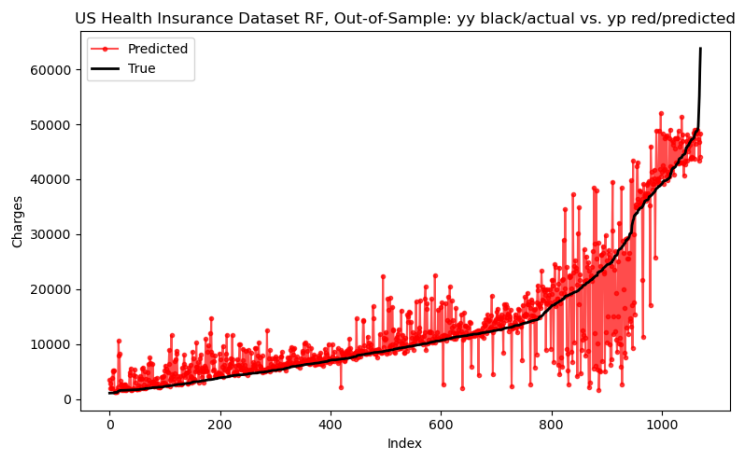


Figure 47: Insurance Charges Random Forrest Out-of-Sample Plot
black: actual y-values, red: predicted y-values

Random Forest Regression metrics: MAE: 2956.7923371477714 RMSE: 5056.020242853598 R2: 0.8225083593592852

Ytest values: rSq: 0.8225083593592852, sst: 154251421569.71732, sse: 27378337885.571674, mse: 25563340.696145356, rmse: 5056.020242853598, mae: 2956.7923371477714, m: 1071

Hyper Parameter tuning info:

Best params:

n estimators: 700, min samples split: 2, min samples leaf: 1, max features: 0.5, max depth: 10, bootstrap: False

Tuned metrics:

Tuned RF MAE: 3087.9428686341853

Tuned RF RMSE: 5141.698517661322

Tuned RF R2: 0.8164419175052696

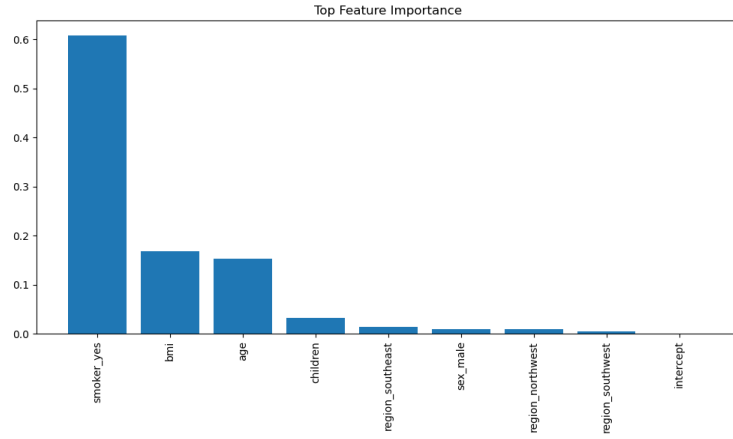


Figure 48: Insurance Charges Feature Importance

3.13 Model Comparison

Table 21: Insurance Charges In-Sample QoF Comparison

Metric	Reg	Ridge	Lasso	Sqrt	Log1p	Box-Cox	Order 2	NN 2L	NN 3L	NN 4L
rSq	0.7509	0.7509	0.7509	0.7527	0.5228	0.7527	0.8477	0.7703	0.8535	0.8674
rSqBar	0.7494	0.7495	0.7496	0.7512	0.5199	0.7512	0.8406	0.7691	0.8527	0.8667
sst	1.961e+11	1.961e+11	1.961e+11	1.961e+11	1.961e+11	1.961e+11	1.961e+11	1.961e+11	1.961e+11	1.961e+11
sse	4.884e+10	4.885e+10	4.884e+10	4.85e+10	9.357e+10	4.85e+10	2.986e+10	4.504e+10	2.873e+10	2.6e+10
sde	6062	6061	6060	6041	8391	6041	4835	5819	4648	4421
mse0	3.65e+07	3.651e+07	3.65e+07	3.625e+07	6.993e+07	3.625e+07	2.232e+07	3.366e+07	2.147e+07	1.943e+07
rmse	6042	6042	6042	6020	8363	6020	4724	5802	4634	4408
mae	4171	4182	4170	3614	4220	3614	2833	3996	2814	2557
smape	37.81	37.68	37.72	27.69	26.29	27.69	26.43	35.78	27.01	25.04
m	1338	1338	1338	1338	1338	1338	1338	1338	1338	1338
dfr	9	8	8	9	9	9	61	8	8	8
df	1329	1330	1330	1329	1329	1329	1277	1330	1330	1330
fStat	445.2	501	501.2	449.3	161.8	449.3	116.5	557.5	968.4	1088
aic	2.332e+04	2.331e+04	2.331e+04	2.331e+04	2.419e+04	2.331e+04	2.276e+04	2.321e+04	2.26e+04	2.247e+04
bic	2.336e+04	2.336e+04	2.336e+04	2.335e+04	2.423e+04	2.335e+04	2.308e+04	2.325e+04	2.265e+04	2.251e+04

Table 22: Insurance Charges Out-of-Sample QoF Comparison

Metric	Reg	Ridge	Lasso	Sqrt	Log1p	Box-Cox	Order 2	NN 2L	NN 3L	NN 4L
rSq	0.8	0.7993	0.7997	0.8077	0.5468	0.8077	0.88	0.8211	0.8926	0.884
rSqBar	0.7938	0.7939	0.7943	0.8017	0.5328	0.8017	0.8452	0.8163	0.8897	0.8809
sst	4.265e+10	4.265e+10	4.265e+10	4.265e+10	4.265e+10	4.265e+10	4.265e+10	4.265e+10	4.265e+10	4.265e+10
sse	8.53e+09	8.561e+09	8.541e+09	8.202e+09	1.933e+10	8.202e+09	5.117e+09	7.631e+09	4.58e+09	4.947e+09
sde	5739	5738	5732	5627	8638	5627	4972	5418	4197	4362
mse0	3.183e+07	3.194e+07	3.187e+07	3.06e+07	7.211e+07	3.06e+07	1.909e+07	2.847e+07	1.709e+07	1.846e+07
rmse	5642	5652	5645	5532	8492	5532	4370	5336	4134	4296
mae	3933	3951	3936	3316	4213	3316	2957	3668	2401	2900
smape	35.02	34.84	35.01	26.47	25.08	26.47	28.54	33.65	23.25	28.72
m	268	268	268	268	268	268	268	268	268	268
dfr	9	8	8	9	9	9	61	8	8	8
df	259	260	260	259	259	259	207	260	260	260
fStat	115.1	129.4	129.8	120.9	34.73	120.9	24.89	149.1	270.2	247.7
aic	4648	4647	4646	4637	4867	4637	4615	4616	4479	4500
bic	4680	4676	4675	4670	4899	4670	4834	4645	4508	4529

Table 23: Insurance Charges Cross-Validation QoF Comparison

Metric	Reg	Ridge	Lasso	Sqrt	Log1p	Box-Cox	Order 2	NN 2L	NN 3L	NN 4L
rSq	0.7401	0.7402	0.7401	0.7447	0.5104	0.7447	0.836	0.7589	0.8427	0.842
rSqBar	0.7321	0.7332	0.7331	0.7368	0.4953	0.7368	0.7884	0.7524	0.8384	0.8377
sst	3.902e+10	3.902e+10	3.902e+10	3.902e+10	3.902e+10	3.902e+10	3.902e+10	3.902e+10	3.902e+10	3.902e+10
sse	9.937e+09	9.938e+09	9.938e+09	9.869e+09	1.915e+10	9.869e+09	6.302e+09	9.209e+09	6.03e+09	6.057e+09
sde	6188	6177	6176	6160	8578	6160	5492	5942	4793	4802
mse0	3.714e+07	3.714e+07	3.714e+07	3.689e+07	7.159e+07	3.689e+07	2.356e+07	3.442e+07	2.254e+07	2.264e+07
rmse	6083	6084	6083	6055	8433	6055	4826	5852	4721	4730
mae	4206	4217	4206	3646	4275	3646	2913	4045	2831	2786
smape	38.03	37.88	37.94	27.88	26.53	27.88	26.99	36.06	28.2	27.06
m	267.6	267.6	267.6	267.6	267.6	267.6	267.6	267.6	267.6	267.6
dfr	9	8	8	9	9	9	61	8	8	8
df	258.6	259.6	259.6	258.6	258.6	258.6	206.6	259.6	259.6	259.6
fStat	86.05	97.06	97.17	86.85	30.29	86.85	18.49	108.3	186.8	186.6
aic	4680	4678	4678	4677	4854	4677	4658	4657	4541	4542
bic	4713	4707	4707	4710	4887	4710	4877	4686	4570	4570

4 Insurance Annual Premium

4.1 Exploratory Data Analysis

The Insurance Annual Premium dataset contains 100,000 observations each having 54 variables. Some of the variables are:

- age: The age of the primary beneficiary.
- sex: The insurance contractor sex: female or male.
- bmi: The body mass index of the primary beneficiary.
- dependents: The number of dependents covered by health insurance
- smoker: If the primary beneficiary smokes.
- region: The beneficiary's residential area in the US.
- annual_premium: (The target variable) Individual medical costs billed by health insurance.

The Insurance Annual Premium dataset contains 30,083 missing values, so we remove these rows. For modeling purposes, we converted all the categorical variables into dummy variables.

Univariate distributions are visualized in Figure 49 (histograms/count-plots) and Figure 50 (box plots for continuous variables).

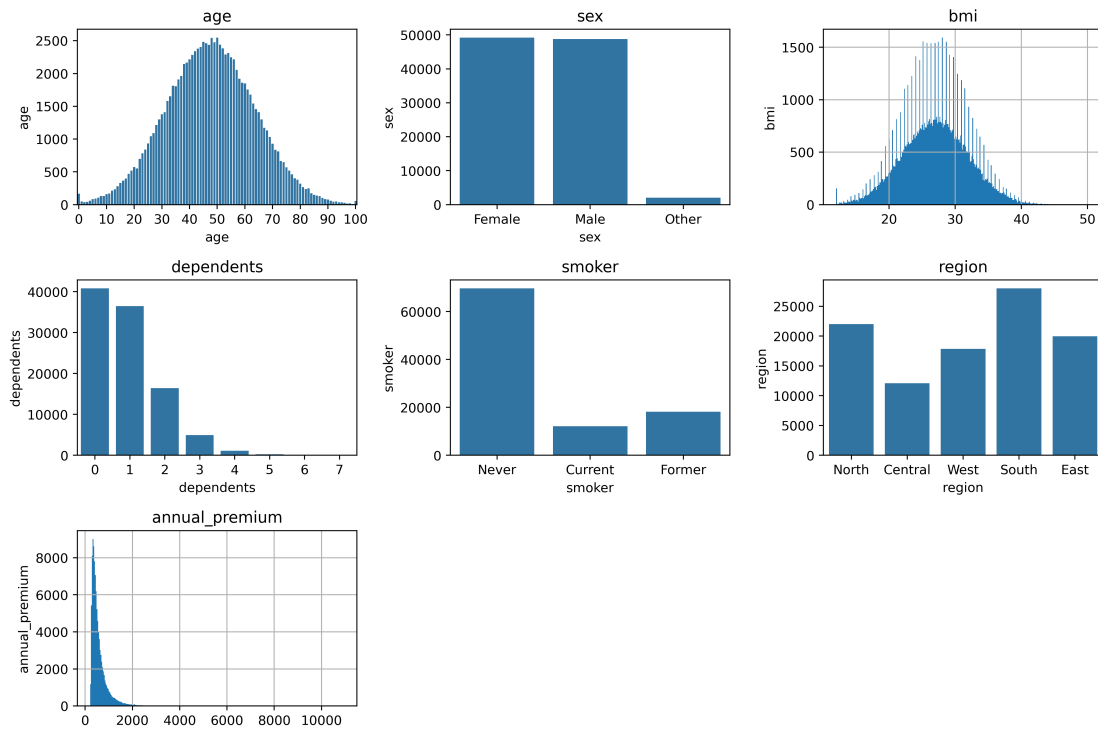


Figure 49: Insurance Annual Premium Histograms and Count-Plots

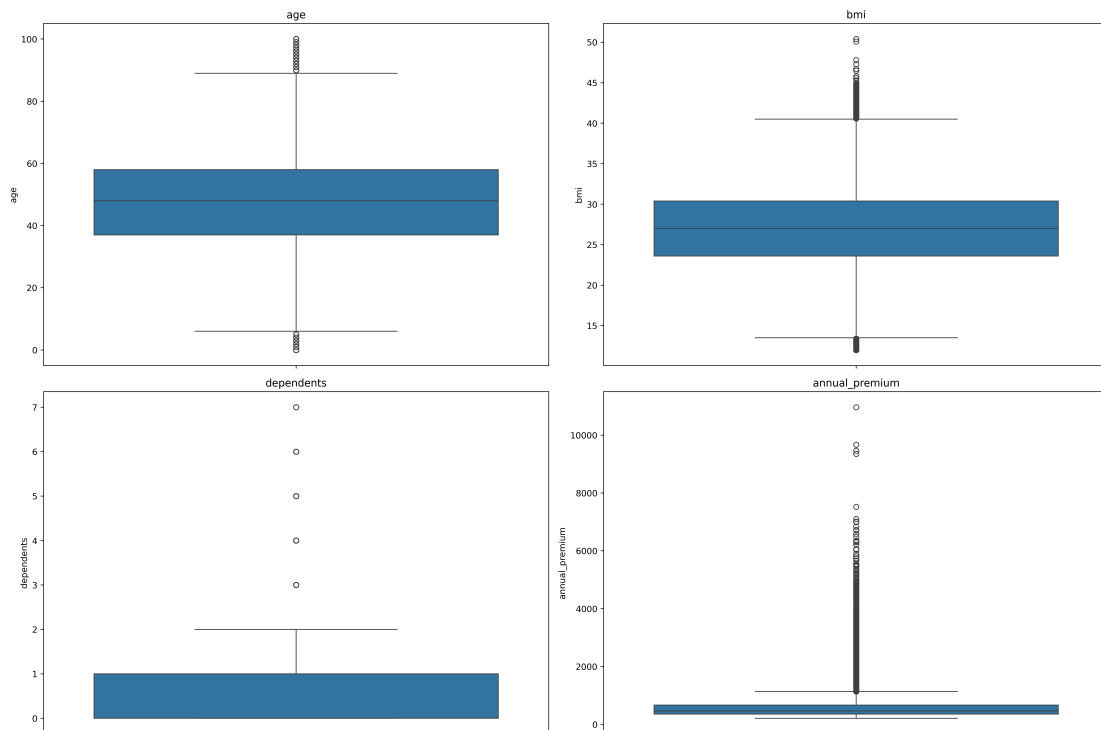


Figure 50: Insurance Annual Premium Box Plots

Figure 51 displays the bivariate relationships between some of the predictors and the target variable, Annual Premium, while the corresponding correlation matrix is provided in Figure 52.

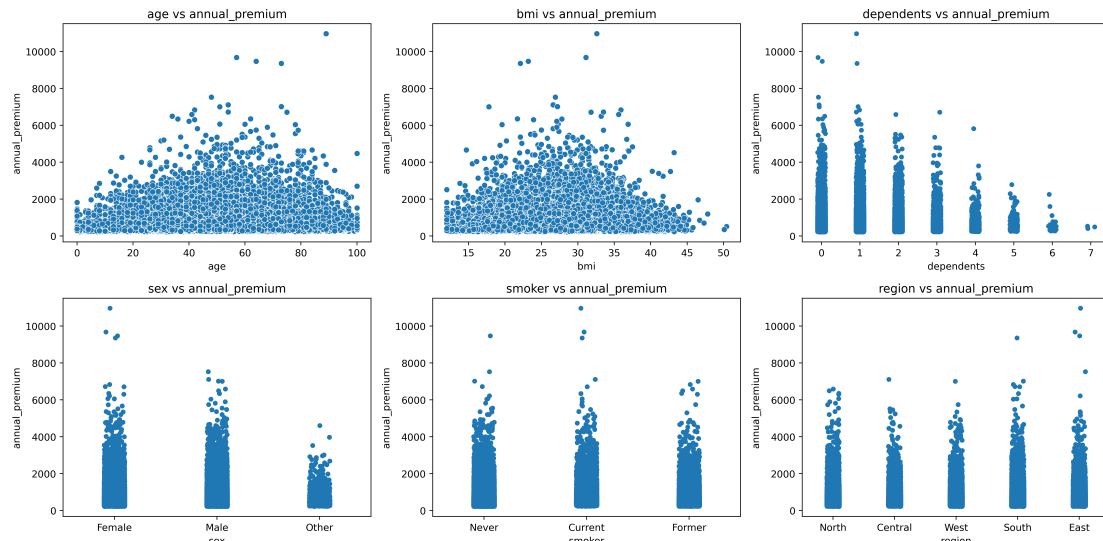


Figure 51: Insurance Annual Premium Scatterplots

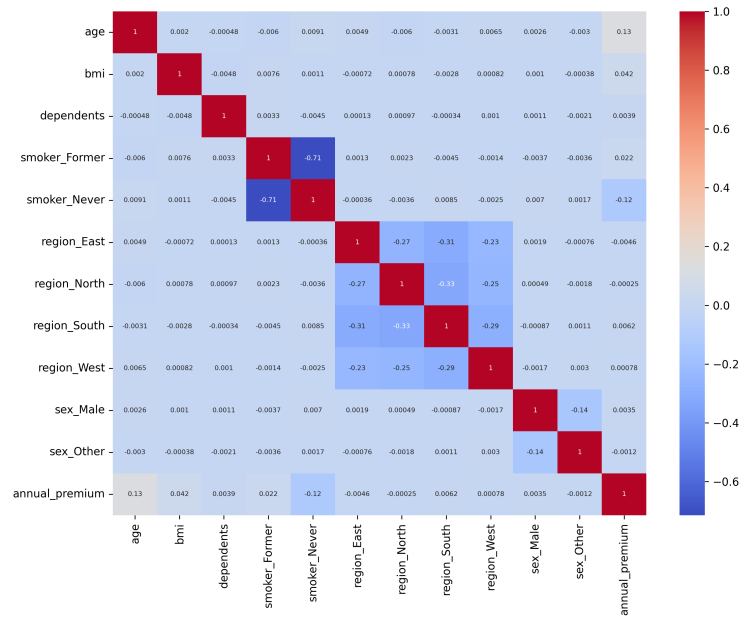


Figure 52: Insurance Annual Premium Correlation Heat Map

Figure 53 illustrates the bivariate relationships among some of the predictor variables, with data points colored by the target variable, Annual Premium.

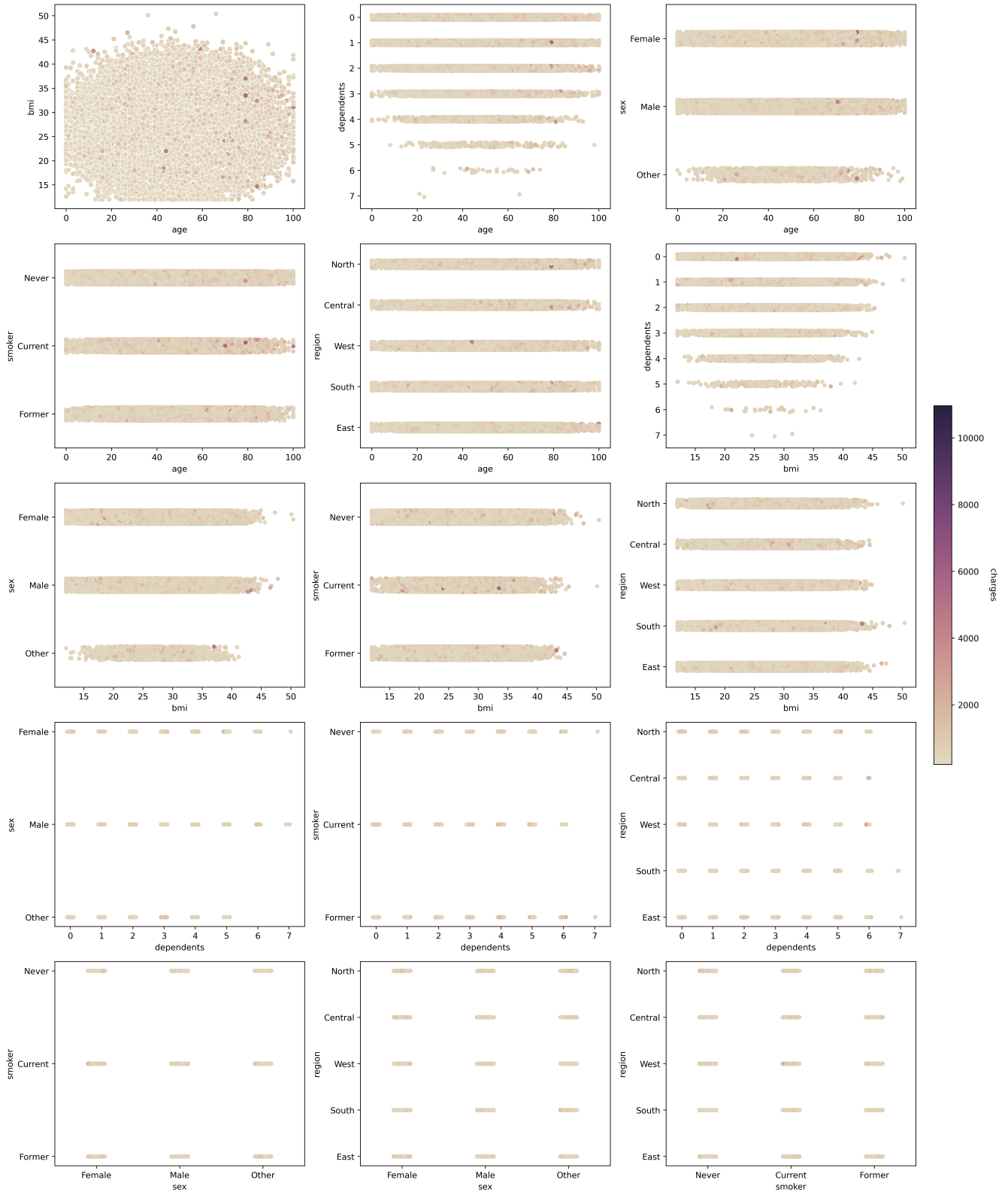


Figure 53: Insurance Annual Premium Hue Plots

4.2 Random Forest

The Random Forest model was trained and evaluated on both in-sample and out-of-sample data. Hyperparameter tuning was performed using cross-validation on the training set to determine the optimal parameters. The tuning process considered the

following parameters:

```
params = {  
    'n_estimators': [200, 300, 400, 500, 700],  
    'max_depth': [10, 20, 30, None],  
    'min_samples_split': [2, 5, 10],  
    'min_samples_leaf': [1, 2, 4],  
    'max_features': ['sqrt', 'log2', 0.5],  
    'bootstrap': [True, False]  
}
```

4.2.1 In-Sample Quality of Fit Metrics

Table 24: In-sample performance metrics of the Random Forest model.

Metric	Value
R^2	0.9549
SST	9.78×10^{11}
SSE	4.41×10^{10}
MSE	441081.88
RMSE	664.14
MAE	374.50
m	100000

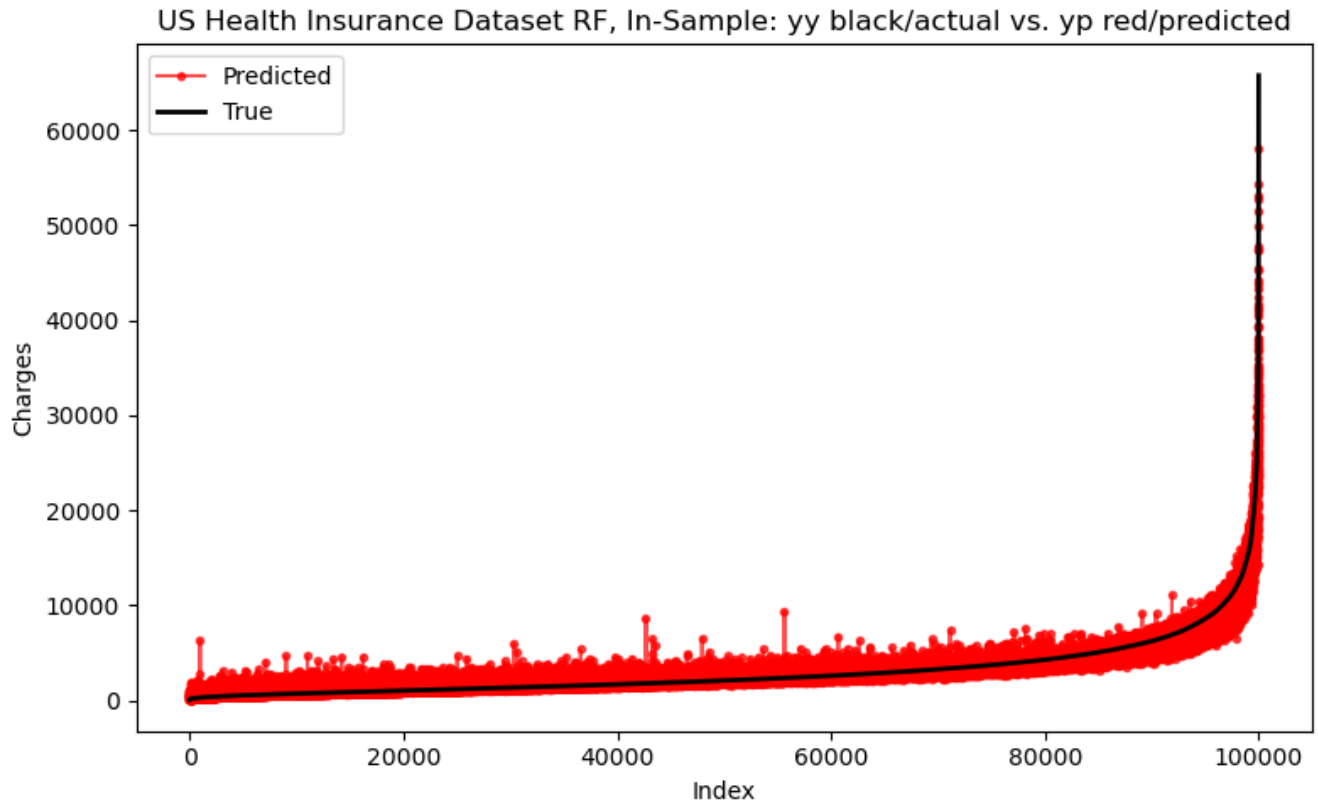


Figure 54: In-sample performance of the Random Forest model on Dataset 2.

5 Out-of-Sample Quality of Fit Metrics

Table 25: Out-of-sample performance metrics of the Random Forest model.

Metric	Value
R^2	0.6510
SST	1.97×10^{11}
SSE	6.87×10^{10}
MSE	3434098.68
RMSE	1853.13
MAE	1025.39
m	20000

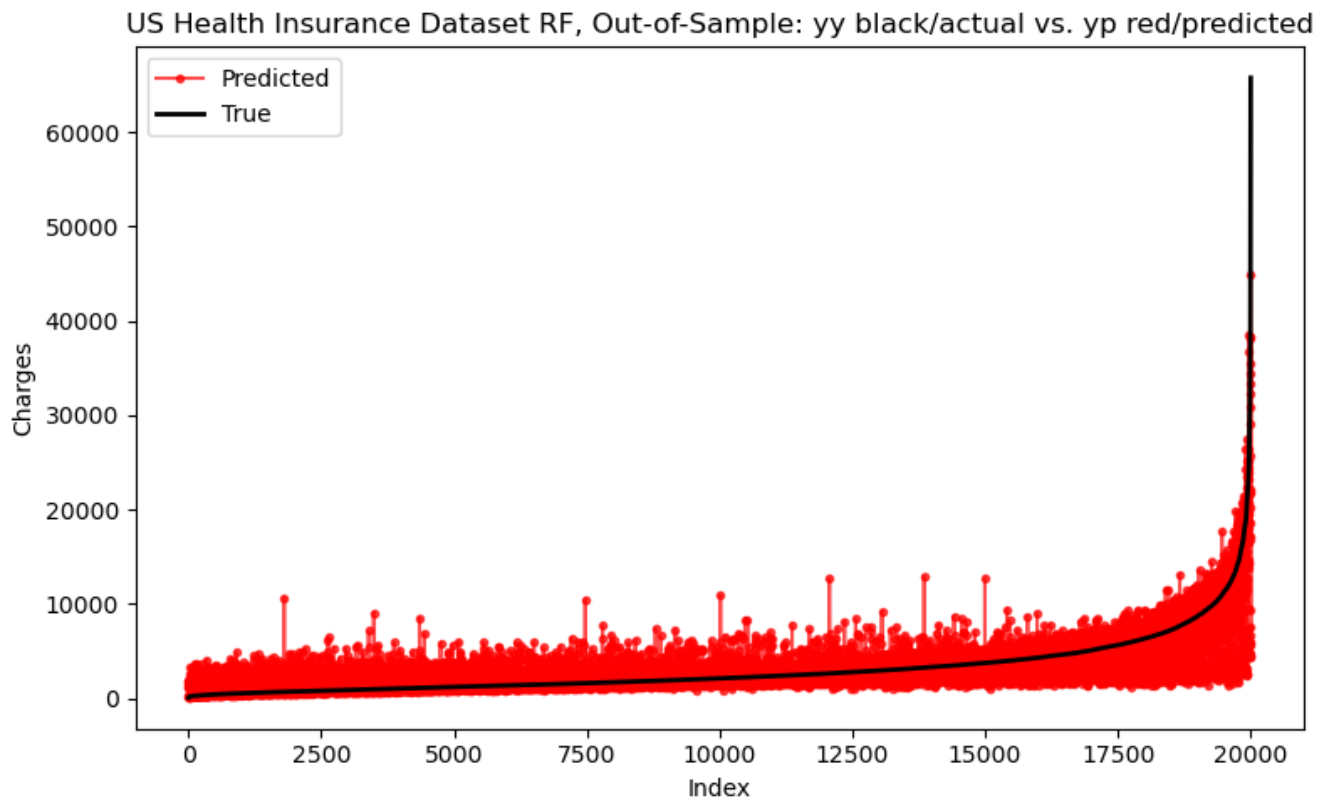


Figure 55: Out-of-sample performance of the Random Forest model on Dataset 2.

5.0.1 Fine-Tuned Quality of Fit Metrics

A total of 20 hyperparameter configurations were randomly sampled, and each was evaluated using 5-fold cross-validation. The following parameter settings were identified as optimal:

Best params: {'n_estimators': 300,
'min_samples_split': 5, 'min_samples_leaf': 4,
'max_features': 0.5, 'max_depth': None,
'bootstrap': False}

Table 26: Fine-tuned performance metrics of the Random Forest model.

Metric	Value
R^2	0.6542
SST	1.97×10^{11}
SSE	6.80×10^{10}
MSE	3402240.89
RMSE	1844.52
MAE	999.62
m	20000

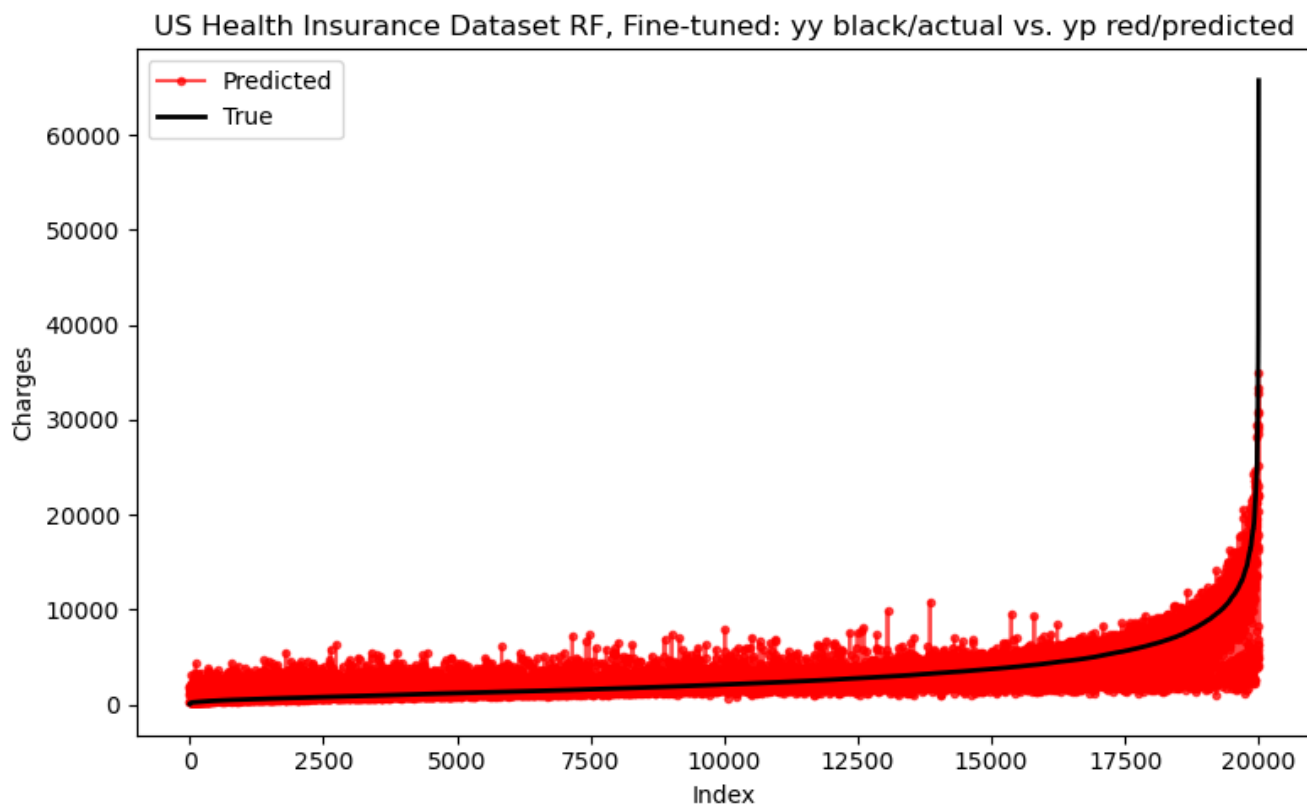


Figure 56: Fine-tuned performance of the Random Forest model on Dataset 2.

5.0.2 Feature Importance from the Best-Performing Model

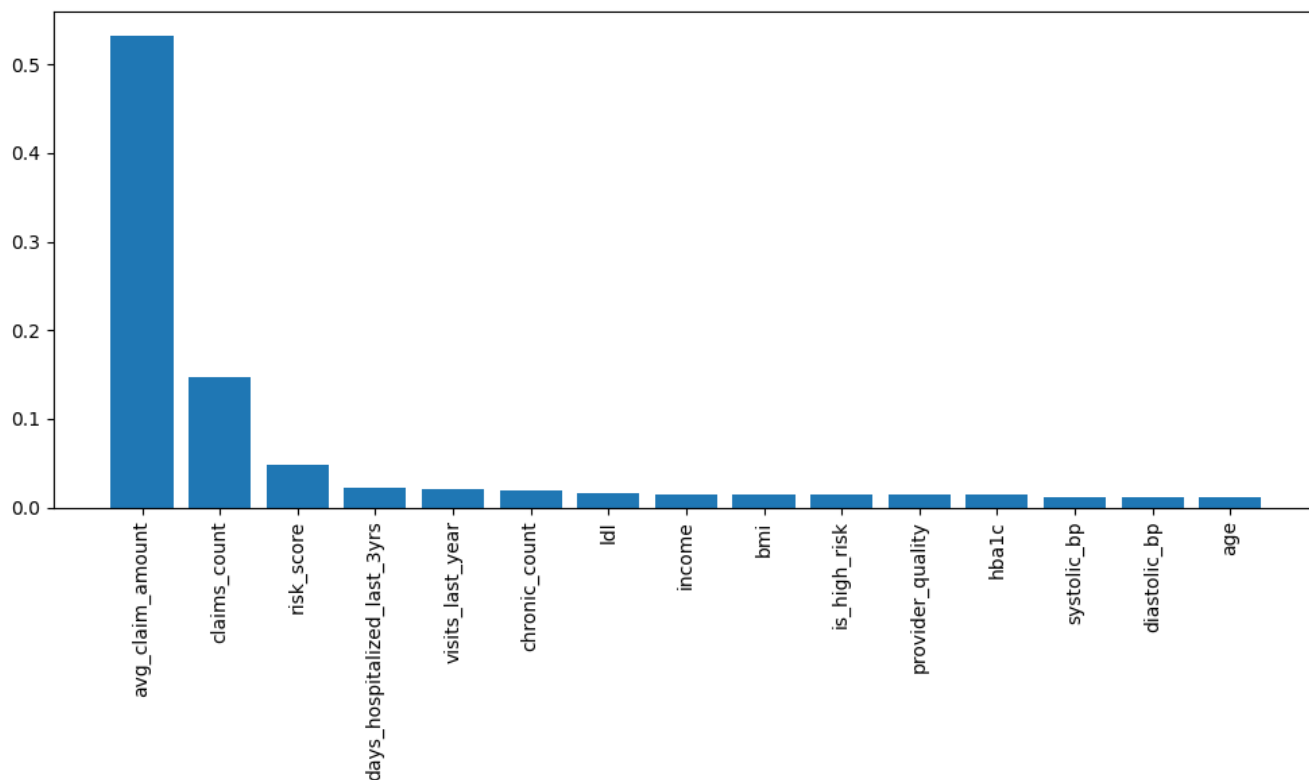


Figure 57: Top 15 Most Important Features.

5.1 Other Modeling Techniques

Alternative approaches (including Ridge and Lasso regression, second-order polynomials, square root transformations, and neural networks) were evaluated but yielded poor predictive performance ($CV R^2 < 0.1$). This underperformance likely stems from the weak correlation between the predictors and the target variable, Annual Premium. Furthermore, the target distribution is heavily skewed, with most values concentrated below \$2,000. Unlike the other regression models tested, Random Forest effectively managed this skewed distribution to provide more reliable results.